

Annual Report to the Nation on the Status of Cancer, Part I: National Cancer Statistics

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BACKGROUND: The American Cancer Society, the Centers for Disease Control and Prevention, the National Cancer Institute, and the North American Association of Central Cancer Registries collaborate to provide annual updates on cancer occurrence and trends in the United States. **METHODS:** Data on new cancer diagnoses during 2001 through 2016 were obtained from the Centers for Disease Control and Prevention-funded and National Cancer Institute-funded population-based cancer registry programs and compiled by the North American Association of Central Cancer Registries. Data on cancer deaths during 2001 through 2017 were obtained from the National Center for Health Statistics' National Vital Statistics System. Trends in incidence and death rates for all cancers combined and for the leading cancer types by sex, racial/ethnic group, and age were estimated by joinpoint analysis and characterized by the average annual percent change during the most recent 5 years (2012-2016 for incidence and 2013-2017 for mortality). **RESULTS:** Overall, cancer incidence rates decreased 0.6% on average per year during 2012 through 2016, but trends differed by sex, racial/ethnic group, and cancer type. Among males, cancer incidence rates were stable overall and among non-Hispanic white males but decreased in other racial/ethnic groups; rates increased for 5 of the 17 most common cancers, were stable for 7 cancers (including prostate), and decreased for 5 cancers (including lung and bronchus [lung] and colorectal). Among females, cancer incidence rates increased during 2012 to 2016 in all racial/ethnic groups, increasing on average 0.2% per year; rates increased for 8 of the 18 most common cancers (including breast), were stable for 6 cancers (including colorectal), and decreased for 4 cancers (including lung). Overall, cancer death rates decreased 1.5% on average per year during 2013 to 2017, decreasing 1.8% per year among males and 1.4% per year among females. During 2013 to 2017, cancer death rates decreased for all cancers combined among both males and females in each racial/ethnic group, for 11 of the 19 most common cancers among males (including lung and colorectal), and for 14 of the 20 most common cancers among females (including lung, colorectal, and breast). The largest declines in death rates were observed for melanoma of the skin (decreasing 6.1% per year among males and 6.3% among females) and lung (decreasing 4.8% per year among males and 3.7% among females). Among children younger than 15 years, cancer incidence rates increased an average of 0.8% per year during 2012 to 2016, and cancer death rates decreased an average of 1.4% per year during 2013 to 2017. Among adolescents and young adults aged 15 to 39 years, cancer incidence rates increased an average of 0.9% per year during 2012 to 2016, and cancer death rates decreased an average of 1.0% per year during 2013 to 2017. **CONCLUSIONS:** Although overall cancer death rates continue to decline, incidence rates are leveling off among males and are increasing slightly among females. These trends reflect population changes in cancer risk factors, screening test use, diagnostic practices, and treatment advances. Many cancers can be prevented or treated effectively if they are found early. Population-based cancer incidence and mortality data can be used to inform efforts to decrease the cancer burden in the United States and regularly monitor progress toward goals. *Cancer* 2020;126:2225-2249. © 2020 American Cancer Society.

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INTRODUCTION

Each year, the Centers for Disease Control and Prevention (CDC), the American Cancer Society (ACS), the National Cancer Institute (NCI), and the North American Association of Central Cancer Registries (NAACCR) collaborate to provide updated information about cancer occurrence and trends in the United States and to address a special topic of interest.¹ Part I of this report focuses on national cancer statistics, and part II characterizes progress in achieving Healthy People 2020 objectives related to four common cancers.²

MATERIALS AND METHODS

Data Sources

Cancer incidence data

Population-based cancer incidence data by age, sex, and race/ethnicity were obtained from registries that participate in the CDC's National Program of Cancer Registries (NPCR) and/or the NCI's Surveillance, Epidemiology, and End Results (SEER) Program.³ Data from registries that satisfied NAACCR's criteria for data quality and completeness were included in the analyses; an exception was that data from New Mexico did not meet NAACCR completeness criteria but, after review, were included in this report.⁴ For rate analyses, 49 states and 1 territory (Puerto Rico) met data criteria for every year during 2012 through 2016, whereas, for trend analyses, 46 states met data criteria for every year during 2001 through 2016, representing 99% and 93% of the population of the United States and Puerto Rico, respectively.

Anatomic site and histology were coded according to *International Classification of Diseases for Oncology, third edition* (ICD-O-3)⁵ and categorized according to SEER site groups.⁶ Only cases defined as malignant were included in this report, except that in situ and malignant bladder cancers were combined when reporting bladder cancer incidence rates.

Cancer mortality data

Cause of death was based on death certificate information reported to state vital statistics offices and compiled into a national file through CDC's National Center for Health Statistics' (NCHS) National Vital Statistics System.⁷ The underlying causes of death were selected according to the ICD tenth revision, then categorized according to SEER site groups to maximize comparability with ICD-O classifications.⁶

Population data

Population estimates used as rate denominators were a modification of intercensal (for July 1, 2001-2009) and vintage 2017 (for July 1, 2010-2017) annual county population estimates by age, sex, bridged-race, and Hispanic origin produced by the US Census Bureau's Population Estimates Program, in collaboration with NCHS, and with support from the NCI.⁸ The estimates incorporate bridged, single-race estimates derived from the original multiple-race categories in the 2000 and 2010 Censuses and permit estimation of race-specific statistics.⁹

Demographic characteristics

Rates and trends are presented by sex, racial/ethnic group, and age (all ages, children aged 0-14 years, and adolescents and young adults [AYAs] aged 15-39 years). Information about race and ethnicity was collected separately and was based on information abstracted from medical records or death certificates. In this report, information about race and ethnicity was combined to create 5 mutually exclusive racial/ethnic groups: non-Hispanic white, non-Hispanic black, non-Hispanic American Indian/Alaska Native (AI/AN), non-Hispanic Asian/Pacific Islander (API), and Hispanic (any race). For purposes of brevity, in this report, the non-Hispanic groups will be referred to by race only. To minimize racial misclassification for AI/AN, this group was restricted to residents in Indian Health Service Purchased/Referred Care Delivery Areas (PRCDA)¹⁰ in states that provided county-level information (ie, excluding Minnesota). Persons with other or unknown race or unknown ethnicity were included in overall rates but were not included as separate categories.

For males and females of all ages, rates and trends are presented for each cancer site that was 1 of the 15 most common sites for incidence or death in any racial/ethnic group. For children and AYAs, rates and trends are presented for each cancer site that was 1 of the 3 most common sites for incidence or death in any racial/ethnic group.

Statistical Methods

Cross-sectional incidence (2012-2016) and death (2013-2017) rates for all ages combined, children, and AYAs by cancer site and racial/ethnic group were calculated using SEER*Stat software, version 8.3.5.¹¹ All rates were age-standardized to the 2000 US standard population and were expressed per 100,000 standard population. Incidence rates were adjusted for potential

delays in case reporting.¹² Corresponding 95% confidence intervals (CIs) were calculated as modified γ intervals and allow for informal comparisons between groups, without specifying a referent group. Rates based on fewer than 16 cases were deemed to be statistically unreliable and were suppressed.

Temporal trends in delay-adjusted cancer incidence (2001–2016) and death (2001–2017) rates were estimated using joinpoint regression.¹³ The maximum number of joinpoints was based on the length of the period, with 2 joinpoints (3 line segments) allowed for the 16-year period for incidence and a maximum of 3 joinpoints (4 line segments) allowed for the 17-year period for deaths. Trends were characterized by the annual percent change (APC), the slope of a single segment, and the average APC (AAPC), which is a summary measure over a fixed interval.¹³ To determine whether APCs were significantly different from zero, a t test was used; to determine whether AAPCs were significantly different from zero, a t test was used for 0 joinpoints, and a z test was used for ≥ 1 joinpoint. Rates were considered to increase if the slope of the trends (APC or AAPC) was greater than zero ($P < .05$) and to decrease if it was less than zero ($P < .05$); otherwise, rates were considered stable. Trends based on fewer than 10 cases in any of the data years were considered statistically unreliable and were suppressed. All statistical tests were 2-sided. Corresponding 95% CIs for trends were calculated using the parametric method and allow for informal comparisons between groups, without specifying a referent group.

RESULTS

Cancer Incidence Rates and Trends

Overall, cancer incidence rates decreased 0.6% on average per year during the most recent 5 years (2012–2016), but trends differed by sex, being stable among males (AAPC, -1.0% ; $P = .14$) and increasing slightly among females (AAPC, 0.2% ; $P = .007$) (Table 1). The overall incidence rate of cancer among males decreased during 2001 through 2013 but was stable during 2013 through 2016, which resulted in a stable trend for the most recent 5 years (Fig. 1 and Table 2). Among females, overall cancer incidence rates were stable during 2001 through 2003 and increased slightly during 2003 through 2016 (APC, 0.2%) (Fig. 1).

By racial/ethnic group, overall cancer incidence rates in the most recent 5 years were stable only among white males and decreased among black, API, AI/AN,

and Hispanic males (Table 1). Overall cancer incidence rates in the most recent 5 years increased among females in every racial/ethnic group (Table 1). The overall cancer incidence rate was similar among white persons and black persons but was higher among black males than white males and higher among white females than black females (Fig. 2). The overall cancer incidence rate was higher among males than females in every racial/ethnic group, except APIs, in which the rates were similar.

The stable trend in overall cancer incidence among males was largely driven by stable trends for 7 of the 17 most common cancers: myeloma, thyroid, leukemia, esophagus, non-Hodgkin lymphoma, stomach, and prostate (Fig. 3A and Table 1). Incidence rates decreased for 5 cancers among males: lung and bronchus (lung) (AAPC, -2.6%), larynx (-2.3%), urinary bladder (bladder) (-1.6%), colon and rectum (colorectal) (-1.1%), and brain and other nervous systems (ONS) (-0.3%). However, incidence rates increased for 5 cancers: liver and intrahepatic bile duct (liver) (AAPC, 2.5%), melanoma of the skin (melanoma) (2.2%), kidney and renal pelvis (kidney) (1.5%), oral cavity and pharynx (1.2%), and pancreas (1.1%). Trends for the 3 most common cancers differed by racial/ethnic group: prostate cancer incidence rates decreased among Hispanic males but were stable among white, black, API, and AI/AN males, and lung and colorectal cancer incidence rates decreased among white, black, API, and Hispanic males but were stable among AI/AN males (Table 1).

The slight increase in overall incidence among females was in part because incidence rates increased for 8 of the 18 most common cancers (liver [AAPC, 3.7%], melanoma [1.9%], corpus and uterus, not otherwise specified [uterus] [1.3%], myeloma [1.2%], pancreas [1.0%], kidney [0.6%], breast [0.5%], and oral cavity and pharynx [0.5%]) but decreased for only 4 cancers (ovary [AAPC, -1.6%], lung [-1.1%], bladder [-0.9%], and brain and ONS [-0.5%]) and were stable for 6 cancers (cervix, stomach, leukemia, non-Hodgkin lymphoma, thyroid, and colorectum) (Fig. 3B and Table 1). Breast cancer incidence rates increased among females in every racial/ethnic group, whereas lung cancer incidence rates decreased among white, black, and Hispanic females but were stable among API and AI/AN females. Colorectal cancer incidence rates decreased among white, API, and AI/AN females but were stable among black and Hispanic females (Table 1).

Overall cancer incidence rates in 2012 through 2016 ranged from 363 to 517 cases per 100,000 persons

TABLE 1. Age-Standardized, Delay-Adjusted Incidence Rates^a and Fixed-Interval Trends (2012–2016)^b for the Most Common Cancers,^c All Ages, by Sex, Age Group, and Racial/Ethnic Group^d for Areas in the United States With High-Quality Incidence Data^e

Sex/Cancer Site or Type	Rank	All Races			Non-Hispanic White			Non-Hispanic Black			Non-Hispanic API			Non-Hispanic AI/AN (PRCDA)			Hispanic		
		Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P
All sites																			
Both sexes		447.9 (447.6, 448.2)	-0.6 (-0.7, -0.5) ^f	< .001	467.5 (467.1, 467.9)	-0.4 (-0.5, -0.3) ^f	< .001	486.9 (465.8, 467.9)	-1.1 (-1.4, -0.8) ^f	< .001	293.6 (292.5, 294.8)	-0.5 (-0.6, -0.4) ^f	< .001	401.4 (396.7, 406.1)	-0.1 (-0.3, 0.1)	.40	348.9 (348.0, 349.7)	-0.8 (-0.9, -0.7) ^f	< .001
Males		487.9 (487.4, 488.4)	-1.0 (-2.4, 0.3)	.14	503.0 (502.4, 503.6)	-0.8 (-2.2, 0.7)	.31	547.6 (545.8, 549.3)	-1.7 (-2.8, -0.6) ^f	.003	296.5 (294.7, 298.3)	-1.5 (-1.7, -1.3) ^f	.003	420.0 (412.6, 427.5)	-0.7 (-1.1, -0.3) ^f	.001	377.8 (376.4, 379.2)	-1.4 (-2.1, -0.7) ^f	.001
Females		421.4 (421.0, 421.8)	0.2 (0.1, 0.3) ^f	.007	444.5 (444.0, 445.0)	0.3 (0.2, 0.4) ^f	< .001	412.8 (411.6, 414.1)	0.3 (0.2, 0.4) ^f	< .001	295.7 (294.1, 297.3)	0.4 (0.3, 0.6) ^f	< .001	391.9 (385.8, 398.2)	0.5 (0.3, 0.7) ^f	< .001	333.6 (332.5, 334.8)	0.6 (0.3, 0.9) ^f	.002
Children (ages 0–14 y)		16.8 (16.7, 17.0)	0.8 (0.7, 1.0) ^f	< .001	17.8 (17.6, 18.1)	0.9 (0.7, 1.0) ^f	< .001	13.5 (13.1, 13.8)	1.2 (0.8, 1.6) ^f	< .001	14.7 (14.1, 15.3)	1.8 (1.1, 2.5) ^f	< .001	12.4 (11.2, 13.8)	-0.2 (-1.6, 1.3)	.81	16.3 (16.0, 16.6)	0.5 (0.2, 0.8) ^f	.004
AYAs (ages 15–39 y)		75.5 (75.2, 75.7)	0.9 (0.8, 1.0) ^f	< .001	84.1 (83.8, 84.5)	1.1 (1.0, 1.2) ^f	< .001	63.5 (62.9, 64.1)	0.7 (0.6, 0.9) ^f	< .001	54.8 (54.0, 55.6)	1.8 (1.5, 2.0) ^f	< .001	61.9 (59.4, 64.6)	0.8 (0.1, 1.4) ^f	.02	62.0 (61.5, 62.5)	2.4 (1.6, 3.2) ^f	< .001
Males																			
Prostate	1	108.1 (107.9, 108.3)	-2.1 (-2.2, 5.5)	.58	101.7 (101.5, 102.0)	-1.9 (-10.0, 6.9)	.66	182.3 (181.3, 183.3)	-2.1 (-6.3, 2.4)	.36	56.3 (55.5, 57.1)	-3.0 (-9.1, 3.6)	.37	75.7 (72.5, 79.0)	-2.3 (-7.8, 3.4)	.42	98.2 (97.5, 99.0)	-4.0 (-7.6, -0.3) ^f	.04
Lung and bronchus	2	69.5 (69.3, 69.7)	-2.6 (-2.7, -2.4) ^f	< .001	73.4 (73.2, 73.6)	-2.3 (-2.5, -2.2) ^f	< .001	83.9 (83.2, 84.6)	-3.2 (-3.6, -2.7) ^f	< .001	44.4 (43.7, 45.1)	-1.6 (-1.8, -1.3) ^f	< .001	62.9 (60.0, 65.9)	-5.7 (-12.1, 1.3)	.11	36.2 (35.7, 36.6)	-2.8 (-3.0, -2.5) ^f	< .001
Colon and rectum	3	45.1 (45.0, 45.3)	-1.1 (-2.1, -0.2) ^f	.03	44.7 (44.5, 44.9)	-1.2 (-1.9, -0.6) ^f	.002	55.0 (54.4, 55.6)	-2.7 (-3.0, -2.4) ^f	< .001	36.2 (35.5, 36.8)	-2.3 (-2.5, -2.0) ^f	< .001	51.9 (49.3, 54.6)	-0.6 (-1.2, 0.0)	.06	42.9 (42.4, 43.4)	-2.2 (-2.5, -1.9) ^f	< .001
Urinary bladder	4	35.5 (35.4, 35.7)	-1.6 (-2.3, -0.8) ^f	< .001	40.1 (40.0, 40.3)	-1.4 (-2.3, -0.5) ^f	.003	20.4 (20.1, 20.8)	-1.2 (-2.9, 0.7)	.21	15.3 (14.9, 15.8)	-0.6 (-1.0, -0.3) ^f	.001	20.7 (18.9, 22.6)	0.9 (-0.1, 2.0)	.08	19.0 (18.7, 19.4)	-1.5 (-1.8, -1.2) ^f	< .001
Melanoma of the skin	5	28.5 (28.3, 28.6)	2.2 (2.0, 2.4) ^f	< .001	36.4 (36.2, 36.5)	2.6 (2.4, 2.8) ^f	< .001	1.2 (1.1, 1.3)	0.0 (-1.0, 1.1)	.93	1.6 (1.4, 1.7)	-0.5 (-1.2, 0.3)	.18	10.8 (9.6, 12.1)	7.9 (4.9, 11.0) ^f	< .001	5.1 (5.0, 5.3)	0.5 (0.0, 1.1)	.052
Non-Hodgkin lymphoma	6	23.8 (23.6, 23.9)	-1.0 (-2.3, 0.2)	.11	25.1 (25.0, 25.2)	-0.9 (-2.1, 0.3)	.14	17.9 (17.5, 18.2)	0.3 (0.0, 0.5) ^f	.03	16.4 (16.0, 16.8)	0.4 (0.0, 0.9) ^f	.04	16.6 (15.2, 18.2)	-0.8 (-2.2, 0.6)	.24	20.1 (19.8, 20.4)	0.0 (-0.4, 0.4)	.91
Kidney and renal pelvis	7	22.9 (22.8, 23.0)	1.5 (1.3, 1.7) ^f	< .001	23.4 (23.3, 23.5)	1.6 (1.3, 2.0) ^f	< .001	26.6 (26.2, 26.9)	1.3 (0.7, 1.8) ^f	< .001	11.4 (11.1, 11.8)	0.7 (-0.6, 2.0)	.28	31.8 (29.9, 33.8)	0.7 (-0.4, 1.9)	.19	20.8 (20.5, 21.2)	2.2 (1.4, 3.1) ^f	< .001
Leukemia	8	19.4 (19.3, 19.5)	-0.1 (-1.5, 1.4)	.94	20.7 (20.6, 20.9)	0.2 (-1.6, 1.9)	.86	15.1 (14.8, 15.5)	1.1 (0.6, 1.6) ^f	< .001	10.3 (9.9, 10.6)	0.8 (0.2, 1.4) ^f	.02	14.2 (12.8, 15.6)	1.0 (-0.4, 2.4)	.15	14.2 (13.9, 14.5)	0.7 (0.2, 1.2) ^f	.005
Oral cavity and pharynx	9	18.1 (18.0, 18.2)	1.2 (1.0, 1.4) ^f	< .001	20.0 (19.9, 20.1)	1.9 (1.6, 2.1) ^f	< .001	14.6 (14.4, 14.9)	-1.8 (-2.2, -1.5) ^f	< .001	11.6 (11.3, 12.0)	0.7 (0.2, 1.2) ^f	.006	15.2 (13.9, 16.5)	1.3 (-0.1, 2.7)	.06	11.3 (11.0, 11.5)	-0.6 (-1.0, -0.1) ^f	.02
Pancreas	10	14.7 (14.6, 14.8)	1.1 (1.0, 1.1) ^f	< .001	14.9 (14.8, 15.0)	1.2 (1.1, 1.4) ^f	< .001	17.8 (17.4, 18.1)	0.6 (0.3, 1.0) ^f	.001	10.3 (10.0, 10.7)	0.6 (0.2, 1.0) ^f	.01	13.2 (11.9, 14.6)	2.2 (0.4, 4.0) ^f	.02	11.9 (11.7, 12.2)	0.5 (0.1, 0.8) ^f	.01
Liver and intrahepatic bile duct	11	13.0 (12.9, 13.0)	2.5 (1.9, 3.1) ^f	< .001	10.8 (10.7, 10.9)	2.1 (0.8, 3.5) ^f	.005	18.4 (18.1, 18.7)	1.9 (0.7, 3.1) ^f	.004	20.0 (19.6, 20.5)	-0.9 (-1.3, -0.6) ^f	< .001	22.6 (21.1, 24.2)	4.7 (3.4, 6.1) ^f	< .001	19.6 (19.3, 19.9)	0.4 (-0.6, 1.4)	.44
Stomach	12	9.1 (9.0, 9.1)	-1.5 (-3.0, 0.1)	.06	7.7 (7.7, 7.8)	-1.6 (-2.5, -0.8) ^f	.002	14.1 (13.8, 14.4)	-1.9 (-2.1, -1.6) ^f	< .001	13.6 (13.2, 14.0)	-2.8 (-3.3, -2.4) ^f	< .001	12.0 (10.7, 13.3)	-3 (-4.0, -1.9) ^f	< .001	12.1 (11.8, 12.3)	-2.0 (-2.4, -1.7) ^f	< .001
Myeloma	13	9.0 (9.0, 9.1)	1.1 (-0.1, 2.4)	.07	8.3 (8.2, 8.3)	1.1 (-0.1, 2.4)	.07	17.7 (17.4, 18.1)	2.1 (1.8, 2.5) ^f	< .001	5.2 (4.9, 5.4)	-1.5 (-4.0, 1.1)	.27	9.2 (8.0, 10.4)	2.3 (-9.0, 14.9)	.71	8.4 (8.2, 8.6)	-0.6 (-3.2, 2.1)	.62
Esophagus	14	7.9 (7.9, 8.0)	-0.5 (-1.3, 0.3)	.20	8.7 (8.7, 8.8)	-0.6 (-1.0, -0.2) ^f	.005	6.6 (6.5, 6.8)	-4.6 (-5.0, -4.3) ^f	< .001	3.6 (3.4, 3.8)	-0.9 (-2.0, 0.1)	.08	7.9 (7.0, 9.0)	-0.2 (-2.3, 2.0)	.86	5.0 (4.8, 5.1)	-1.7 (-2.3, -1.2) ^f	< .001

TABLE 1. Continued

Sex/Cancer Site or Type	Rank	All Races			Non-Hispanic White			Non-Hispanic Black			Non-Hispanic API			Non-Hispanic AI/AN (PRCDA)			Hispanic		
		Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P
Brain and other nervous system	15	7.8 (7.7, 7.8)	-0.3 (-0.4, -0.2) ^f	<.001	8.9 (8.8, 8.9)	0.0 (-0.1, 0.1)	.78	5.0 (4.9, 5.2)	0.3 (-0.3, 0.8)	.32	4.4 (4.2, 4.6)	-0.5 (-1.7, 0.8)	.40	5.1 (4.4, 5.9)	-0.3 (-2.4, 1.8)	.74	5.7 (5.6, 5.9)	-0.9 (-1.1, -0.7) ^f	<.001
Thyroid	16	7.5 (7.5, 7.6)	0.5 (-1.6, 2.6)	.65	8.4 (8.3, 8.4)	0.6 (-1.5, 2.6)	.58	4.0 (3.9, 4.2)	1.8 (-0.5, 4.1)	.11	7.4 (7.1, 7.7)	0.5 (-3.4, 4.5)	.82	5.0 (4.3, 5.8)	4.8 (2.3, 7.3) ^f	.001	6.2 (6.1, 6.4)	2.4 (0.8, 4.0) ^f	.007
Larynx	17	5.8 (5.7, 5.8)	-2.3 (-2.5, -2.2) ^f	<.001	5.9 (5.8, 5.9)	-1.9 (-2.1, -1.8) ^f	<.001	8.4 (8.1, 8.6)	-3.1 (-3.5, -2.8) ^f	<.001	2.0 (1.9, 2.2)	-2.7 (-4.1, -1.3) ^f	.001	5.3 (4.5, 6.1)	-1.5 (-3.1, 0.1)	.07	4.8 (4.6, 4.9)	-3.2 (-3.7, -2.8) ^f	<.001
Females																			
Breast	1	126.8 (126.5, 127.0)	0.5 (0.3, 0.6) ^f	<.001	133.0 (132.7, 133.3)	0.6 (0.4, 0.7) ^f	<.001	129.6 (128.9, 130.3)	0.6 (0.3, 0.9) ^f	.003	96.5 (95.6, 97.4)	1.5 (1.2, 1.9) ^f	<.001	102.5 (99.4, 105.6)	0.8 (0.3, 1.2) ^f	.004	96.1 (95.5, 96.6)	0.3 (0.1, 0.6) ^f	.01
Lung and bronchus	2	51.8 (51.7, 51.9)	-1.1 (-1.3, -1.0) ^f	<.001	57.5 (57.3, 57.7)	-0.9 (-1.0, -0.7) ^f	<.001	49.3 (48.9, 49.8)	-1.6 (-2.0, -1.1) ^f	<.001	28.2 (27.7, 28.7)	-1.5 (-3.1, 0.2)	.08	51.2 (48.9, 53.5)	-0.3 (-0.8, 0.2)	.22	23.1 (22.8, 23.4)	-0.8 (-1.1, -0.6) ^f	<.001
Colon and rectum	3	34.4 (34.3, 34.6)	-0.3 (-0.7, 0.1)	.15	34.4 (34.3, 34.6)	-0.5 (-0.7, -0.3) ^f	.001	40.8 (40.4, 41.2)	-1.3 (-2.6, 0.0)	.06	26.3 (25.8, 26.8)	-2.4 (-2.7, -2.1) ^f	<.001	41.8 (39.8, 43.9)	-0.7 (-1.2, -0.2) ^f	.01	30.0 (29.7, 30.3)	0.3 (-0.9, 1.6)	.54
Corpus and uterus, NOS	4	26.9 (26.8, 27.1)	1.3 (1.2, 1.4) ^f	<.001	27.6 (27.5, 27.8)	1.1 (1.0, 1.3) ^f	<.001	27.1 (26.8, 27.4)	2.4 (2.2, 2.6) ^f	<.001	19.4 (19.0, 19.8)	2.2 (1.9, 2.5) ^f	<.001	24.5 (23.1, 26.0)	1.7 (0.7, 2.6) ^f	.002	24.0 (23.8, 24.3)	2.9 (2.4, 3.4) ^f	<.001
Thyroid	5	22.0 (21.9, 22.1)	-0.2 (-1.3, 0.8)	.65	23.3 (23.2, 23.5)	-0.3 (-1.3, 0.7)	.50	14.2 (14.0, 14.5)	-0.1 (-2.3, 2.2)	.94	22.4 (22.0, 22.8)	-1.8 (-4.0, 0.5)	.11	16.9 (15.7, 18.1)	5.0 (3.9, 6.1) ^f	<.001	23.3 (23.1, 23.6)	1.7 (0.5, 2.9) ^f	.01
Melanoma of the skin	6	17.6 (17.5, 17.6)	1.9 (1.5, 2.2) ^f	<.001	23.8 (23.7, 23.9)	2.5 (2.1, 2.8) ^f	<.001	0.9 (0.9, 1.0)	-0.5 (-1.5, 0.4)	.25	1.2 (1.1, 1.3)	-0.3 (-1.6, 0.9)	.58	6.9 (6.0, 7.7)	2.1 (0.6, 3.6) ^f	.008	4.4 (4.3, 4.5)	0.5 (-0.2, 1.2)	.14
Non-Hodgkin lymphoma	7	16.4 (16.3, 16.5)	-0.1 (-0.2, 0.0)	.16	17.1 (17.0, 17.2)	-0.1 (-0.2, 0.1)	.23	12.7 (12.5, 13.0)	0.5 (0.3, 0.8) ^f	.001	11.2 (10.9, 11.5)	0.3 (-0.2, 0.9)	.23	14.2 (13.0, 15.4)	-0.7 (-1.7, 0.3)	.14	15.4 (15.2, 15.7)	0.3 (0.0, 0.6)	.07
Kidney and renal pelvis	8	11.7 (11.7, 11.8)	0.6 (0.3, 0.9) ^f	.002	11.8 (11.7, 11.9)	1.2 (0.1, 2.2) ^f	.03	13.6 (13.3, 13.8)	0.5 (-0.3, 1.3)	.17	5.3 (5.1, 5.5)	1.1 (0.2, 2.0) ^f	.02	18.8 (17.5, 20.1)	1.5 (0.4, 2.7) ^f	.01	11.8 (11.6, 12.0)	1.6 (1.2, 1.9) ^f	<.001
Leukemia	9	11.7 (11.6, 11.8)	0.0 (-1.3, 1.3)	.98	12.3 (12.3, 12.4)	0.0 (-1.5, 1.6)	.95	9.7 (9.5, 9.9)	0.4 (-1.8, 2.7)	.69	6.6 (6.4, 6.8)	1.0 (0.5, 1.6) ^f	.001	8.7 (7.8, 9.7)	0.4 (-1.3, 2.1)	.65	10.0 (9.8, 10.2)	1.4 (0.9, 1.9) ^f	<.001
Ovary	10	11.4 (11.3, 11.4)	-1.6 (-1.7, -1.4) ^f	<.001	11.9 (11.8, 12.0)	-1.7 (-1.8, -1.5) ^f	<.001	9.4 (9.2, 9.6)	-0.8 (-1.1, -0.4) ^f	<.001	9.5 (9.2, 9.8)	-0.2 (-0.6, 0.2)	.36	11.2 (10.2, 12.3)	-1.2 (-2.9, 0.5)	.14	10.2 (10.0, 10.4)	-1.3 (-1.6, -1.0) ^f	<.001
Pancreas	11	11.3 (11.2, 11.4)	1.0 (0.9, 1.2) ^f	<.001	11.2 (11.1, 11.2)	1.2 (1.0, 1.3) ^f	<.001	14.9 (14.6, 15.1)	0.6 (0.3, 1.0) ^f	.001	8.8 (8.6, 9.1)	0.7 (0.2, 1.2) ^f	.007	10.7 (9.7, 11.8)	0.5 (-0.9, 1.9)	.47	10.0 (9.8, 10.2)	0.6 (0.4, 0.9) ^f	<.001
Urinary bladder	12	8.8 (8.7, 8.8)	-0.9 (-1.1, -0.7) ^f	<.001	9.8 (9.8, 9.9)	-0.7 (-0.8, -0.5) ^f	<.001	6.8 (6.6, 6.9)	-0.4 (-0.7, 0.0) ^f	.03	3.7 (3.6, 3.9)	-0.8 (-1.7, 0.2)	.11	5.9 (5.1, 6.7)	1.3 (-0.5, 3.1)	.15	5.0 (4.8, 5.1)	-1.3 (-1.9, -0.8) ^f	<.001
Cervix	13	7.7 (7.7, 7.8)	1.0 (-0.3, 2.4)	.15	7.2 (7.1, 7.3)	1.4 (-0.1, 3.0)	.07	9.3 (9.1, 9.5)	-2.3 (-2.7, -1.9) ^f	<.001	6.2 (5.9, 6.4)	-1.9 (-2.5, -1.3) ^f	<.001	9.4 (8.5, 10.3)	-0.2 (-1.2, 0.8)	.70	10.0 (9.8, 10.1)	1.1 (-0.4, 2.7)	.14
Oral cavity and pharynx	14	6.5 (6.5, 6.6)	0.5 (0.3, 0.7) ^f	<.001	7.1 (7.0, 7.2)	0.9 (0.7, 1.1) ^f	<.001	5.3 (5.1, 5.4)	-0.7 (-0.9, -0.4) ^f	<.001	5.4 (5.2, 5.6)	0.2 (-0.6, 1.0)	.68	5.9 (5.2, 6.7)	-0.3 (-2.2, 1.7)	.75	4.3 (4.2, 4.5)	0.3 (-0.3, 1.0)	.25
Myeloma	15	5.9 (5.8, 5.9)	1.2 (0.4, 2.0) ^f	.01	5.0 (4.9, 5.0)	1.0 (0.0, 2.0) ^f	.0496	13.1 (12.9, 13.4)	2.3 (2.0, 2.7) ^f	<.001	3.2 (3.0, 3.4)	0.9 (0.0, 1.7) ^f	.045	6.3 (5.5, 7.1)	0.2 (-1.8, 2.2)	.87	5.8 (5.7, 6.0)	2.0 (1.0, 2.9) ^f	.001
Brain and other nervous system	16	5.6 (5.6, 5.7)	-0.5 (-0.8, -0.3) ^f	.001	6.4 (6.3, 6.5)	-0.3 (-0.6, 0.0)	.07	3.6 (3.5, 3.8)	-0.2 (-0.6, 0.3)	.48	3.3 (3.1, 3.5)	-1.7 (-6.3, 3.1)	.47	3.3 (2.8, 3.9)	-4.4 (-6.9, -1.8) ^f	.005	4.5 (4.4, 4.6)	-0.7 (-1.1, -0.3) ^f	.003
Stomach	17	4.7 (4.7, 4.7)	0.1 (-0.4, 0.6)	.61	3.5 (3.5, 3.6)	0.0 (-0.6, 0.6)	.98	7.6 (7.4, 7.8)	-1.2 (-1.6, -0.9) ^f	<.001	8.0 (7.8, 8.3)	-2.5 (-3.1, -2.0) ^f	<.001	6.9 (6.1, 7.7)	-1.5 (-3.2, 0.3)	.09	7.7 (7.5, 7.8)	-1.3 (-1.7, -0.9) ^f	<.001

TABLE 1. Continued

Sex/Cancer Site or Type	Rank	All Races			Non-Hispanic White			Non-Hispanic Black			Non-Hispanic API			Non-Hispanic AI/AN (PRCDA)			Hispanic		
		Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P
Liver and intrahepatic bile duct	18	4.5 (4.5, 4.6)	3.7 (3.4, 4.0) ^f	< .001	3.8 (3.7, 3.8)	4.0 (3.7, 4.2) ^f	< .001	5.6 (5.4, 5.7)	3.7 (3.1, 4.2) ^f	< .001	7.5 (7.3, 7.8)	-1.0 (-1.5, -0.4) ^f	.003	9.8 (8.9, 10.8)	2.9 (1.3, 4.5) ^f	.001	7.5 (7.3, 7.7)	2.3 (1.8, 2.8) ^f	< .001
Children																			
Leukemia	1	5.2 (5.1, 5.3)	0.9 (0.5, 1.2) ^f	< .001	5.3 (5.2, 5.4)	0.8 (0.5, 1.1) ^f	< .001	3.3 (3.1, 3.5)	1.6 (1.0, 2.3) ^f	< .001	5.2 (4.9, 5.6)	0.6 (-0.2, 1.5)	.15	4.7 (3.9, 5.6)	15.3 (0.6, 32.0) ^f	.04	6.2 (6.0, 6.4)	0.8 (0.2, 1.4) ^f	.01
Brain and other nervous system	2	3.8 (3.7, 3.8)	0.8 (0.5, 1.1) ^f	< .001	4.3 (4.2, 4.4)	1.0 (0.6, 1.4) ^f	< .001	3.0 (2.8, 3.1)	1.5 (0.7, 2.3) ^f	.001	2.8 (2.5, 3.0)	1.1 (-0.1, 2.4)	.07	2.4 (1.9, 3.1)	— ^g		3.0 (2.9, 3.1)	0.0 (-0.4, 0.4)	.99
Lymphoma	3	1.6 (1.6, 1.7)	0.9 (0.4, 1.4) ^f	.001	1.7 (1.6, 1.8)	5.2 (0.7, 9.8) ^f	.03	1.6 (1.5, 1.7)	1.6 (0.5, 2.6) ^f	.005	1.4 (1.3, 1.6)	0.5 (-1.0, 2.1)	.50	1.2 (0.8, 1.6)	— ^g		1.5 (1.4, 1.6)	0.3 (-0.4, 0.9)	.38
AYA																			
Female breast	1	22.4 (22.2, 22.6)	1.2 (0.5, 1.8) ^f	.003	23.1 (22.8, 23.3)	0.6 (0.4, 0.8) ^f	< .001	27.1 (26.6, 27.7)	0.5 (0.2, 0.7) ^f	.002	19.8 (19.2, 20.5)	0.9 (0.4, 1.5) ^f	.003	16.6 (14.6, 18.7)	0.3 (-1.4, 2.2)	.69	17.6 (17.2, 18.0)	2.0 (-0.1, 4.2)	.06
Thyroid	2	12.2 (12.1, 12.3)	1.7 (0.6, 2.8) ^f	.006	14.0 (13.8, 14.1)	1.4 (0.3, 2.5) ^f	.02	5.8 (5.6, 6.0)	-0.1 (-5.5, 5.5)	.96	12.1 (11.7, 12.5)	0.0 (-3.1, 3.1)	.99	8.6 (7.7, 9.6)	3.7 (2.4, 4.9) ^f	< .001	10.5 (10.3, 10.7)	4.7 (4.2, 5.3) ^f	< .001
Testis	3	10.9 (10.8, 11.1)	0.8 (0.6, 0.9) ^f	< .001	13.5 (13.3, 13.6)	0.6 (0.4, 0.8) ^f	< .001	2.5 (2.4, 2.7)	0.5 (-0.7, 1.8)	.36	4.0 (3.7, 4.3)	3.3 (2.1, 4.5) ^f	< .001	10.8 (9.5, 12.3)	0.6 (-0.7, 2.0)	.32	10.8 (10.5, 11.1)	2.9 (2.4, 3.4) ^f	< .001
Lymphoma	4	7.7 (7.6, 7.8)	-0.4 (-0.5, -0.3) ^f	< .001	8.3 (8.2, 8.4)	-0.4 (-0.5, -0.2) ^f	< .001	8.4 (8.2, 8.6)	-0.3 (-0.6, 0.0)	.06	5.2 (5.0, 5.5)	2.1 (1.4, 2.7) ^f	< .001	4.0 (3.3, 4.6)	-0.4 (-2.6, 2.0)	.75	6.1 (6.0, 6.3)	-0.1 (-0.4, 0.2)	.43

Abbreviations: AAPC, average annual percent change; AI/AN, American Indian/Alaska Native; API, Asian/Pacific Islander; AYAs, adolescents and young adults; CI, confidence interval; NOS, not otherwise specified; PRCDA, Indian Health Services Purchased/Referred Care Delivery Area.

^aRates are per 100,000 persons, adjusted for potential delays in reporting, and age standardized to the 2000 US standard population (19 age groups; US Bureau of the Census. Current Population Reports, Publication 25-1130. US Government Printing Office; 2000 [Census P25-1130]).

^bFixed interval trends are characterized as the AAPC, the weighted average of the annual percent changes over the fixed interval 2012 to 2016, using the underlying Joinpoint model (with up to 2 joinpoints) for the period 2001 to 2016. The Joinpoint Regression Program was used (version 4.7.0.0; Statistical Research and Applications Branch, National Cancer Institute; February 2019).

^cCancers are listed in descending rank order of sex-specific, age-adjusted incidence rates for 2012 to 2016 for all racial/ethnic groups combined. More than 15 cancers may appear for males and females to include the top 15 cancers in each racial/ethnic group.

^dRacial/ethnic groups are mutually exclusive. Data for non-Hispanic AI/ANs are restricted to PRCDA counties (excluding Minnesota).

^eThe following National Program of Cancer Registries (NPCR) and Surveillance, Epidemiology, and End Results (SEER) areas are reported by the North American Association of Central Cancer Registries (NAACCR) as meeting high-quality incidence data standards for the specified time periods. Registries included in the incidence rate analysis (49 states and 1 territory): Alabama, Alaska, Arizona, Arkansas, California, Colorado, Connecticut, Delaware, Florida, Georgia, Hawaii, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maine, Maryland, Massachusetts, Michigan, Minnesota, Mississippi, Missouri, Montana, Nebraska, New Hampshire, New Jersey, New Mexico, New York, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Puerto Rico, Rhode Island, South Carolina, South Dakota, Tennessee, Texas, Utah, Vermont, Virginia, Washington, West Virginia, Wisconsin, Wyoming. Registries included in the Joinpoint regression analysis (46 states): Alabama, Alaska, Arizona, Arkansas, California, Colorado, Connecticut, Delaware, Florida, Georgia, Hawaii, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maine, Maryland, Massachusetts, Michigan, Minnesota, Missouri, Montana, Nebraska, New Hampshire, New Jersey, New Mexico, New York, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Puerto Rico, Rhode Island, South Carolina, South Dakota, Tennessee, Texas, Utah, Vermont, Washington, West Virginia, Wisconsin, and Wyoming.

^fThe AAPC is statistically significantly different from zero ($P < .05$). Two-sided statistical significance for the AAPC was determined using a t test when the AAPC lay entirely within the last joinpoint segment or a z test when the AAPC extended beyond the last joinpoint segment.

^gThe statistic could not be calculated. The AAPC is based on <10 cases for at least 1 year within the time interval.

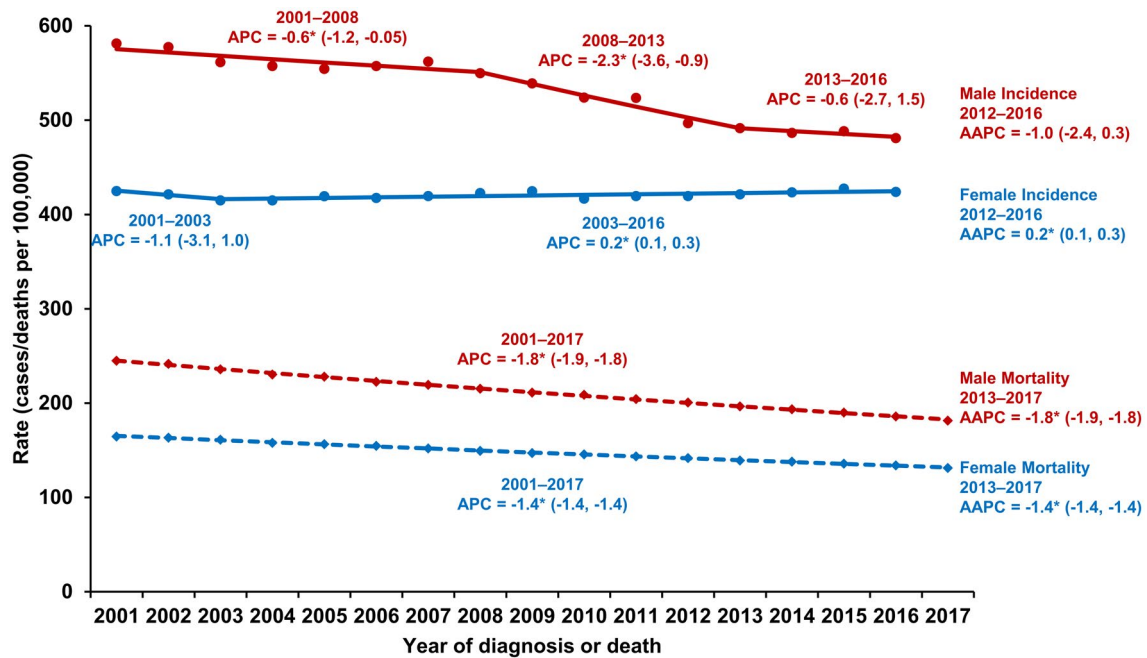


Figure 1. Trends in age-standardized incidence rates (2001–2016) and death rates (2001–2017) are illustrated for all cancer sites combined and for all races and ethnicities combined, by sex. Incidence rates were delay-adjusted and covered 93% of the US population, and death rates covered the entire US population. Trends were estimated using joinpoint regression and characterized by the annual percent change (APC), the slope of a single segment, and the average APC (AAPC), a summary measure of the APCs over a fixed 5-year interval. The 16-year period for incidence allowed up to 3 different APCs, and the 17-year period for mortality allowed up to 4 different APCs. An asterisk (*) indicates that the APC or AAPC was statistically significantly different from zero ($P < .05$). 95% confidence limits are given in parentheses.

(Fig. 4A). During 2012 through 2016, cancer incidence rates decreased in 31 states and were stable in 15 states; trends could not be calculated for 4 states, the District of Columbia, or Puerto Rico because data were not available for the entire period (Fig. 4B).

Cancer Death Rates and Trends

Overall, cancer death rates decreased 1.5% on average per year during 2001 through 2017, decreasing more rapidly among males (APC, -1.8%) than among females (APC, -1.4%) (Fig. 1). Overall cancer death rates decreased during 2013 through 2017 in every racial/ethnic group, decreasing the most among black persons (AAPC, -2.0%) and the least among AI/AN persons (AAPC, -0.6%) (Table 3). The overall cancer death rate was highest among black persons, next highest among AI/AN and white persons, and lowest among API persons (Fig. 5).

During 2013 through 2017, death rates among males decreased for 11 of the 19 most common cancers, were stable for 4 cancers (including prostate), and increased for 4 cancers (oral cavity and pharynx; soft tissue, including heart; brain and ONS; and pancreas) (Fig. 6A and Table 3). Death rates among females increased for

cancers of the uterus, liver, brain and ONS, soft tissue, including heart, and pancreas; rates were stable for oral cavity and pharynx cancer and decreased for 14 of the 20 most common cancers (Fig. 6B and Table 3). The largest declines in death rates were observed for melanoma (AAPC, -6.1% among males and -6.3% among females) and lung cancer (AAPC, -4.8% among males and -3.7% among females) (Fig. 6).

The 3 most common cancer deaths among males were from lung, prostate, and colorectal cancer in most racial/ethnic groups, except API males. Among API males, lung cancer was the most common cancer death, followed by cancer of the liver, colorectum, and prostate (Table 3). Lung cancer death rates among males decreased during 2013 through 2017 in each racial/ethnic group, decreasing the most among black males (AAPC, -5.7%) and Hispanic males (-5.4%). Prostate cancer death rates were stable among white and Hispanic males but decreased among black, API, and AI/AN males. Colorectal cancer death rates were stable among AI/AN males but decreased in all other racial/ethnic groups. Pancreas cancer was the fourth most common cancer death among white and black males, and death rates increased among

TABLE 2. Joinpoint Trends^a in Age-Standardized, Delay-Adjusted Incidence Rates for the Most Common Cancers,^b All Races/Ethnicities Combined by Sex and Age Group, for Areas in the United States With High-Quality Incidence Data,^c 2001 to 2016

Sex/Cancer Site or Type	Trend 1			Trend 2			Trend 3			Fixed Interval 2012-2016		
	Years	APC (95% CI)	P	Years	APC (95% CI)	P	Years	APC (95% CI)	P	Years	APC (95% CI) ^d	P
All sites												
Both sexes	2001-2016	-0.6 (-0.7, -0.5) ^e	< .001	2008-2013	-2.3 (-3.6, -0.9) ^e	.004	2013-2016	-0.6 (-2.7, 1.5)	.52	2013-2016	-0.6 (-0.7, -0.5) ^e	< .001
Males	2001-2008	-0.6 (-1.2, 0.0) ^e	.04	2003-2016	0.2 (0.1, 0.3) ^e	.007					-1.0 (-2.4, 0.3)	.14
Females	2001-2003	-1.1 (-3.1, 1.0)	.28								0.2 (0.1, 0.3) ^e	.007
Children (ages 0-14 y)	2001-2016	0.8 (0.7, 1.0) ^e	< .001								0.8 (0.7, 1.0) ^e	< .001
AYAs (ages 15-39 y)	2001-2016	0.9 (0.8, 1.0) ^e	< .001								0.9 (0.8, 1.0) ^e	< .001
Males												
Prostate	2001-2009	-1.6 (-3.2, 0.1)	.06	2009-2014	-7.5 (-12.2, -2.6) ^e	.008	2014-2016	3.7 (-12.4, 22.8)	.64	2014-2016	-2.1 (-9.2, 5.5)	.58
Lung and bronchus	2001-2007	-1.6 (-1.8, -1.3) ^e	< .001	2007-2016	-2.6 (-2.7, -2.4) ^e	< .001					-2.6 (-2.7, -2.4) ^e	< .001
Colon and rectum	2001-2012	-3.2 (-3.4, -3.0) ^e	< .001	2012-2016	-1.1 (-2.1, -0.2) ^e	.03					-1.1 (-2.1, -0.2) ^e	.03
Urinary bladder	2001-2005	0.1 (-0.5, 0.7)	.72	2005-2014	-0.9 (-1.1, -0.7) ^e	< .001	2014-2016	-2.3 (-4.0, -0.5) ^e	.02	2014-2016	-1.6 (-2.3, -0.8) ^e	< .001
Melanoma of the skin	2001-2016	2.2 (2.0, 2.4) ^e	< .001								2.2 (2.0, 2.4) ^e	< .001
Non-Hodgkin lymphoma	2001-2014	0.3 (0.1, 0.4) ^e	.003	2014-2016	-2.3 (-5.1, 0.5)	.10					-1.0 (-2.3, 0.2)	.11
Kidney and renal pelvis	2001-2007	3.0 (2.8, 3.2) ^e	< .001	2007-2011	0.0 (-0.5, 0.5)	.88	2011-2016	1.5 (1.3, 1.7) ^e	< .001	2011-2016	1.5 (1.3, 1.7) ^e	< .001
Leukemia	2001-2007	-0.2 (-0.8, 0.4)	.43	2007-2014	2.1 (1.5, 2.7) ^e	< .001	2014-2016	-2.1 (-5.3, 1.1)	.17	2014-2016	-0.1 (-1.5, 1.4)	.94
Oral cavity and pharynx	2001-2005	-0.1 (-1.3, 1.1)	.87	2005-2016	1.2 (1.0, 1.4) ^e	< .001					1.2 (1.0, 1.4) ^e	< .001
Pancreas	2001-2016	1.1 (1.0, 1.1) ^e	< .001								1.1 (1.0, 1.1) ^e	< .001
Liver and intrahepatic bile duct	2001-2009	4.7 (4.1, 5.3) ^e	< .001	2009-2016	2.5 (1.9, 3.1) ^e	< .001					2.5 (1.9, 3.1) ^e	< .001
Stomach	2001-2006	-2.1 (-3.0, -1.3) ^e	< .001	2006-2014	-0.4 (-0.9, 0.1)	.07	2014-2016	-2.5 (-6.0, 1.1)	.14	2014-2016	-1.5 (-3.0, 0.1)	.06
Myeloma	2001-2007	0.7 (0.2, 1.3)	.02	2007-2014	2.8 (2.3, 3.4) ^e	< .001	2014-2016	-0.5 (-3.3, 2.3)	.67	2014-2016	1.1 (-0.1, 2.4)	.07
Esophagus	2001-2008	0.3 (-0.2, 0.8)	.27	2008-2011	-2.9 (-6.4, 0.7)	.10	2011-2016	-0.5 (-1.3, 0.3)	.20	2011-2016	-0.5 (-1.3, 0.3)	.20
Brain and other nervous system	2001-2016	-0.3 (-0.4, -0.2) ^e	< .001								-0.3 (-0.4, -0.2) ^e	< .001
Thyroid	2001-2009	7.0 (6.3, 7.7) ^e	< .001	2009-2014	2.2 (0.6, 3.8) ^e	.01	2014-2016	-1.2 (-5.7, 3.5)	.56	2014-2016	0.5 (-1.6, 2.6)	.65
Larynx	2001-2016	-2.3 (-2.5, -2.2) ^e	< .001								-2.3 (-2.5, -2.2) ^e	< .001
Females												
Breast	2001-2004	-2.8 (-4.3, -1.3) ^e	.002	2004-2016	0.5 (0.3, 0.6) ^e	< .001					0.5 (0.3, 0.6) ^e	< .001
Lung and bronchus	2001-2006	0.7 (0.2, 1.2) ^e	.01	2006-2016	-1.1 (-1.3, -1.0) ^e	< .001					-1.1 (-1.3, -1.0) ^e	< .001
Colon and rectum	2001-2007	-2.4 (-2.6, -2.2) ^e	< .001	2007-2012	-3.4 (-3.8, -3.0) ^e	< .001	2012-2016	-0.3 (-0.7, 0.1)	.15	2012-2016	-0.3 (-0.7, 0.1)	.15
Corpus and uterus, NOS	2001-2003	-2.1 (-4.4, 0.3)	.09	2003-2016	1.3 (1.2, 1.4) ^e	< .001					1.3 (1.2, 1.4) ^e	< .001
Thyroid	2001-2009	7.3 (6.8, 7.8) ^e	< .001	2009-2013	2.3 (0.7, 4.0) ^e	.01	2013-2016	-1.1 (-2.7, 0.5)	.15	2013-2016	-0.2 (-1.3, 0.8)	.65
Melanoma of the skin	2001-2016	1.9 (1.5, 2.2) ^e	< .001								1.9 (1.5, 2.2) ^e	< .001
Non-Hodgkin lymphoma	2001-2016	-0.1 (-0.2, 0.0)	.16								-0.1 (-0.2, 0.0)	.16
Kidney and renal pelvis	2001-2006	3.8 (2.7, 4.9) ^e	< .001	2006-2016	0.6 (0.3, 0.9) ^e	.002	2011-2016	0.0 (-1.3, 1.3)	.98	2011-2016	0.6 (0.3, 0.9) ^e	.002
Leukemia	2001-2008	0.2 (-0.6, 1.1)	.52	2008-2011	3.6 (-2.4, 10.1)	.21					0.0 (-1.3, 1.3)	.98
Ovary	2001-2016	-1.6 (-1.7, -1.4) ^e	< .001								-1.6 (-1.7, -1.4) ^e	< .001
Pancreas	2001-2016	-0.9 (-1.1, -0.7) ^e	< .001								0.0 (-0.3, 2.4)	.15
Urinary bladder	2001-2016	-0.9 (-1.1, -0.7) ^e	< .001								-0.9 (-1.1, -0.7) ^e	< .001
Cervix	2001-2003	-3.8 (-7.6, 0.1)	.06	2003-2013	-1.1 (-1.5, -0.7) ^e	< .001	2013-2016	1.7 (-0.4, 3.9)	.10	2013-2016	1.0 (-0.3, 2.4)	.15
Oral cavity and pharynx	2001-2016	0.5 (0.3, 0.7) ^e	< .001								0.5 (0.3, 0.7) ^e	< .001
Myeloma	2001-2007	0.1 (-0.6, 0.8)	.65	2007-2011	3.7 (1.7, 5.7) ^e	.002	2011-2016	1.2 (0.4, 2.0) ^e	.01	2011-2016	1.2 (0.4, 2.0) ^e	.01
Brain and other nervous system	2001-2005	0.8 (-0.4, 2.1)	.17	2005-2016	-0.5 (-0.8, -0.3) ^e	.001					-0.5 (-0.8, -0.3) ^e	.001

TABLE 2. Continued

Sex/Cancer Site or Type	Trend 1			Trend 2			Trend 3			Fixed Interval 2012-2016		
	Years	APC (95% CI)	P	Years	APC (95% CI)	P	Years	APC (95% CI)	P	APC (95% CI) ^d	P	
Stomach	2001-2008	-1.1 (-1.7, -0.5) ^e	.002	2008-2016	0.1 (-0.4, 0.6)	.61				0.1 (-0.4, 0.6)	.61	
Liver and intrahepatic bile duct	2001-2016	3.7 (3.4, 4.0) ^e	< .001							3.7 (3.4, 4.0) ^e	< .001	
Children												
Leukemia	2001-2016	0.9 (0.5, 1.2) ^e	< .001							0.9 (0.5, 1.2) ^e	< .001	
Brain and other nervous system	2001-2016	0.8 (0.5, 1.1) ^e	< .001							0.8 (0.5, 1.1) ^e	< .001	
Lymphoma	2001-2016	0.9 (0.4, 1.4) ^e	.001							0.9 (0.4, 1.4) ^e	.001	
AVAs												
Female breast	2001-2010	0.0 (-0.4, 0.3)	.95	2010-2016	1.2 (0.5, 1.8) ^e	.003				1.2 (0.5, 1.8) ^e	.003	
Thyroid	2001-2010	5.8 (5.1, 6.6) ^e	< .001	2010-2016	1.7 (0.6, 2.8) ^e	.006				1.7 (0.6, 2.8) ^e	.006	
Testis	2001-2016	0.8 (0.6, 0.9) ^e	< .001							0.8 (0.6, 0.9) ^e	< .001	
Lymphoma	2001-2016	-0.4 (-0.5, -0.3) ^e	< .001							-0.4 (-0.5, -0.3) ^e	< .001	

Abbreviations: AAPC, average annual percent change; APC, annual percent change; AYAs, adolescents and young adults; CI, confidence interval; NOS, not otherwise specified.

^aJoinpoint models with up to 2 joinpoints are based on rates per 100,000 persons, adjusted for potential delays in reporting and age standardized to the 2000 US standard population (19 age groups; US Bureau of the Census. Current Population Reports, Publication 25-1130. US Government Printing Office, 2000 [Census P25-1130]). The Joinpoint Regression Program was used (version 4.7.0.0; Statistical Research and Applications Branch, National Cancer Institute, February 2019).

^bCancers are listed in descending rank order of sex-specific, age-adjusted incidence rates for 2012 to 2016 for all racial/ethnic groups combined. More than 15 cancers may appear for males and females to include the top 15 cancers in each racial/ethnic group.

^cThe following National Program of Cancer Registries (NPCR) and Surveillance, Epidemiology, and End Results (SEER) areas are reported by the North American Association of Central Cancer Registries (NAACCR) as meeting high-quality incidence data standards for 2001 to 2016 (46 states): Alabama, Alaska, Arizona, Arkansas, California, Colorado, Connecticut, Delaware, Florida, Georgia, Hawaii, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maine, Maryland, Massachusetts, Michigan, Minnesota, Missouri, Montana, Nebraska, New Hampshire, New Jersey, New Mexico, New York, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, South Dakota, Texas, Utah, Vermont, Washington, West Virginia, Wisconsin, and Wyoming.

^dThe AAPC is the weighted average of the APCs over the fixed interval 2012 to 2016 using the underlying Joinpoint model for the period 2001 to 2016.

^eThe APC or AAPC is statistically significantly different from zero ($P < .05$). Two-sided statistical significance for the APC was determined using a t test and, for the AAPC, using a t test when the AAPC lay entirely within the last joinpoint segment or a z test when the AAPC extended beyond the last joinpoint segment.

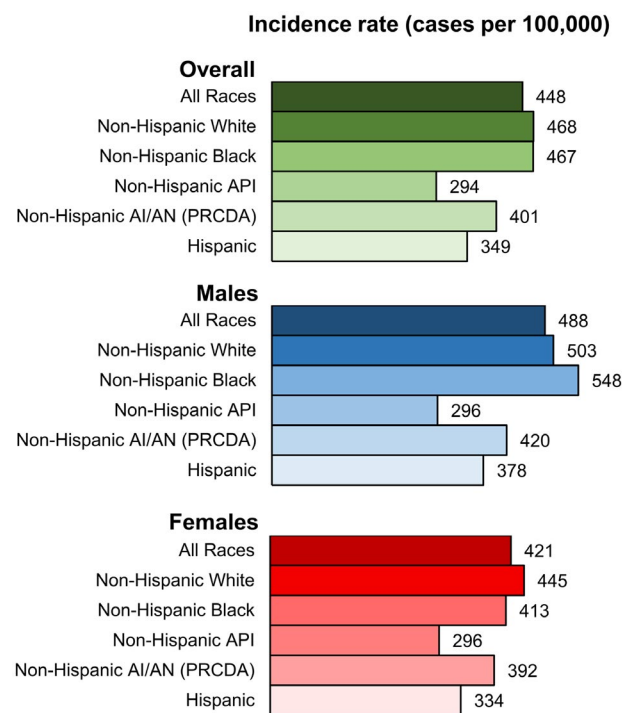


Figure 2. Age-standardized, delay-adjusted overall cancer incidence rates for 2012 through 2016 are illustrated among males and females by racial/ethnic group. Racial/ethnic groups are mutually exclusive. Data for the non-Hispanic American Indian/Alaska Native (AI/AN) population are restricted to Indian Health Service Purchased/Referred Care Delivery Area (PRCDA) counties. API indicates Asian/Pacific Islander.

white males but were stable in other racial/ethnic groups. Liver cancer was the fourth most common cancer death among AI/AN and Hispanic males (and the second most common among API males); death rates increased among white, AI/AN, and Hispanic males, were stable among black males, and decreased among API males.

Among females, the 3 most common cancer deaths were lung, breast, and colorectal, except among Hispanic females, in whom breast cancer was the most common and lung cancer was the second (Table 3). Lung cancer death rates among females decreased in every racial/ethnic group, decreasing the most among black females (−4.6%). Breast and colorectal cancer death rates decreased among white, black, API, and Hispanic females but were stable among AI/AN females. Pancreas cancer was the fourth most common cancer death among females in each racial/ethnic group, and death rates increased among white females, decreased among black females, and were stable in other racial/ethnic groups. Uterine cancer death rates increased among white, black, API, and Hispanic females and could not be calculated among AI/AN females.

Notable changes in trend were observed for melanoma and lung cancer. Melanoma death rates among males were stable during 2009 through 2014 then decreased 7.6% per year during 2014 through 2017, and melanoma death rates among females were stable during 2001 through 2013 then decreased 6.3% per year during 2013 through 2017 (Table 4). In contrast, incidence rates of melanoma increased 2.2% per year among males and 1.9% per year among females during 2001 through 2016 (Table 2). Declines in lung cancer death rates accelerated over the time period studied (Table 4). Among males, lung cancer death rates decreased 2.0% per year during 2001 through 2005, 2.9% during 2005 through 2012, 4.0% during 2012 through 2015, and 5.5% during 2015 through 2017. Among females, lung cancer death rates decreased 0.5% per year during 2001 through 2007, 2.0% during 2007 through 2014, and 4.2% during 2014 through 2017. Lung incidence rates also decreased, but at a slower pace: lung cancer incidence rates among males decreased 2.6% per year during 2007 through 2016 and 1.1% per year among females during 2006 through 2016 (Table 2).

Overall cancer death rates ranged from 125 to 195 deaths per 100,000 standard population (Fig. 7A). During 2013 through 2017, cancer death rates decreased in all states, decreasing 4.3% in Alaska and ≥2% per year in 6 additional states and the District of Columbia (Fig. 7B).

Cancer Among Children

Among children aged birth to 14 years, the incidence rate for all cancer sites combined was 16.8 cases per 100,000 standard population, ranging from 12.4 among AI/AN children to 17.8 among white children (Table 1). Overall cancer incidence rates increased an average of 0.8% per year during 2012 through 2016 (Table 1). The most common cancer types among children included leukemia, brain and ONS, and lymphoma, with increasing trends for each of these cancers during 2001 through 2016 (Table 2). Leukemia rates showed the most variability among racial/ethnic groups, ranging from 3.3 cases per 100,000 standard population among black children to 6.2 among Hispanic children. The cancer death rate among children was 2.1 deaths per 100,000 standard population and was highest among AI/AN children (2.6 deaths per 100,000) and lowest among API children (1.8 deaths per 100,000) (Table 3). Overall cancer death rates among children decreased an average of 1.4% per year during 2013 through 2017. The most common cancer deaths among children were from brain and ONS cancer (0.7 deaths per 100,000) and leukemia (0.5 deaths per 100,000). During

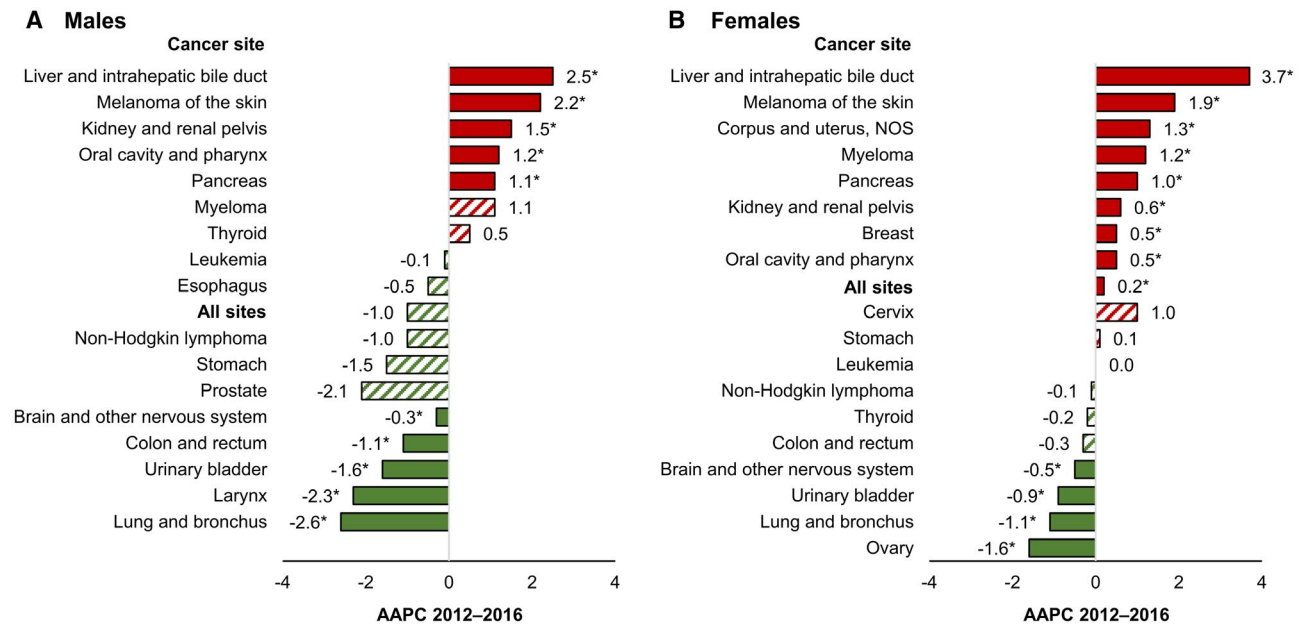


Figure 3. The average annual percent change (AAPC) in age-standardized, delay-adjusted incidence rates for 2012 through 2016 are illustrated for all sites and for the 17 most common cancers in men and the 18 most common cancers in women, all ages, all races/ethnicities combined, among (A) males and (B) females. The AAPC was a weighted average of the annual percent changes (APCs) over the fixed 5-year interval from 2012 to 2016 using the underlying joinpoint regression model, which allowed up to 3 different APCs, for the 16-year period from 2001 to 2016. AAPCs with an asterisk (*) were statistically significantly different from zero ($P < .05$) and are depicted as solid-colored bars; AAPCs in bars with hash marks were not statistically significantly different from zero (stable). NOS indicates not otherwise specified.

2001 through 2017, death rates among children for brain and ONS cancer were stable, whereas death rates from leukemia declined 2.8% per year (Table 4).

Cancer Among Adolescents and Young Adults

Among AYAs aged 15 to 39 years, the incidence rate for all cancer sites combined was 75.5 cases per 100,000 standard population, ranging from 54.8 per 100,000 among API AYAs to 84.1 among white AYAs (Table 1). The most common cancer among AYAs was female breast cancer, which was highest among black AYAs. Incidence rates of thyroid cancer and testicular cancer among AYAs varied by racial/ethnic group and were lowest among black AYAs and highest among white AYAs. Overall cancer incidence rates among AYAs increased an average of 0.9% per year during 2012 through 2016 (Table 1). Three of the 4 most common cancer types among AYAs (female breast, thyroid, and testis) increased during 2012 through 2016, but the fourth (lymphoma) decreased. AYA female breast cancer incidence rates were stable during 2001 through 2010 then increased 1.2% per year during 2010 through 2016 (Table 2). Increasing trends for AYA thyroid cancer

slowed from 5.8% per year during 2001 through 2010 to 1.7% per year during 2010 through 2016 (Table 2).

The cancer death rate among AYAs was 8.9 deaths per 100,000 standard population and was highest among black AYAs (11.4 deaths per 100,000) and AI/AN AYAs (11.0 deaths per 100,000) and lowest among API AYAs (6.9 deaths per 100,000) (Table 3). The most common cancer deaths among AYAs were from female breast cancer (2.2 deaths per 100,000 standard population), brain and ONS cancer (1.0 deaths per 100,000), leukemia (0.9 deaths per 100,000), and colorectal cancer (0.9 deaths per 100,000). Death rates from breast cancer among black AYAs (3.9 deaths per 100,000) were nearly twice as high compared with white AYAs (2.0 deaths per 100,000) and more than twice as high compared with API (1.3 deaths per 100,000), AI/AN (1.6 deaths per 100,000), and Hispanic (1.8 deaths per 100,000) AYAs. Declining trends in overall death rates among AYAs slowed from -3.0% per year during 2001 through 2005 to -1.0% per year during 2005 through 2017 (Table 4). The only notable change in trend in cancer death rates for common sites occurred for AYA female breast cancer, which declined (-3.2% per year)

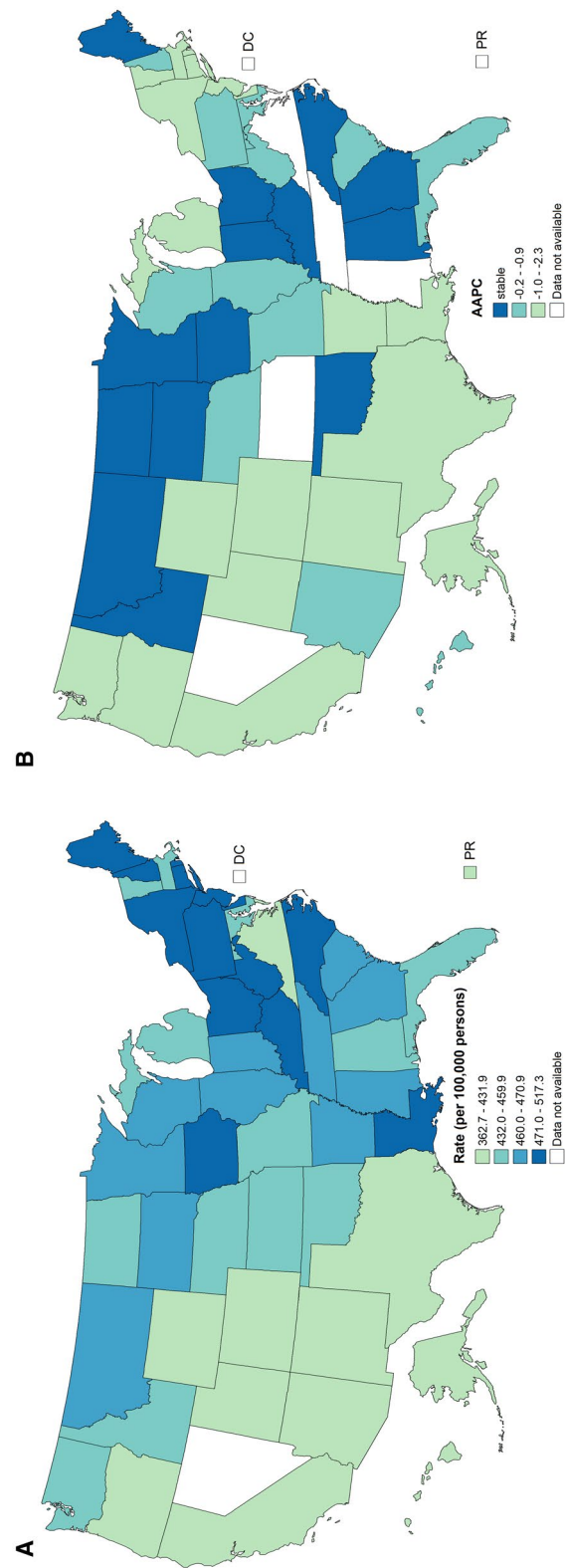


Figure 4. (A) Age-standardized, delay-adjusted overall cancer incidence rates for 2012 through 2016 and (B) the average annual percent change (AAPC) in rates for 2012 through 2016 are illustrated by state. The AAPC was a weighted average of the annual percent changes (APCs) over the fixed 5-year interval from 2012 to 2016 using the underlying joinpoint regression model, which allowed up to 3 different APCs, for the 16-year period from 2001 to 2016. Stable AAPCs were not statistically significantly different from zero ($P < .05$). DC indicates District of Columbia; PR, Puerto Rico.

TABLE 3. Age-Standardized Death Rates^a and Fixed-Interval Trends (2013-2017)^b for the Most Common Causes of Cancer Death,^c by Sex and Racial/Ethnic Group,^d United States^e

Sex/Cancer Site or Type	Rank	All Races			Non-Hispanic White			Non-Hispanic Black			Non-Hispanic API			Non-Hispanic AI/AN (PRCDA)			Hispanic		
		Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P
All sites		158.2 (158.0, 158.3)	-1.5 (-1.6, -1.5) ^f	<.001	162.9 (162.7, 163.1)	-1.4 (-1.4, -1.3) ^f	<.001	186.4 (185.8, 187.1)	-2.0 (-2.1, -2.0) ^f	<.001	98.9 (98.3, 99.6)	-1.3 (-1.4, -1.2) ^f	<.001	166.0 (162.8, 169.3)	-0.6 (-0.9, -0.3) ^f	<.001	111.8 (111.3, 112.3)	-1.3 (-1.4, -1.2) ^f	<.001
Both sexes																			
Males		189.3 (189.0, 189.6)	-1.8 (-1.9, -1.8) ^f	<.001	193.8 (193.4, 194.1)	-1.6 (-1.7, -1.6) ^f	<.001	233.2 (232.1, 234.4)	-2.6 (-2.7, -2.6) ^f	<.001	117.4 (116.2, 118.5)	-1.6 (-1.8, -1.5) ^f	<.001	200.3 (194.8, 205.8)	-0.5 (-1.0, 0.0) ^f	.03	135.6 (134.7, 136.5)	-1.6 (-1.7, -1.6) ^f	<.001
Females		135.5 (135.3, 135.7)	-1.4 (-1.4, -1.4) ^f	<.001	139.9 (139.6, 140.2)	-1.3 (-1.3, -1.3) ^f	<.001	157.5 (156.8, 158.3)	-1.6 (-1.6, -1.5) ^f	<.001	86.7 (84.8, 86.5)	-1.0 (-1.2, -0.9) ^f	<.001	141.0 (137.1, 144.9)	-0.9 (-1.1, -0.6) ^f	<.001	95.1 (94.4, 95.7)	-1.0 (-1.1, -0.9) ^f	<.001
Children (ages 0-14 y)		2.1 (2.0, 2.1)	-1.4 (-1.8, -1.1) ^f	<.001	2.1 (2.0, 2.2)	-1.4 (-1.8, -1.1) ^f	<.001	2.1 (2.0, 2.3)	-1.2 (-1.9, -0.6) ^f	<.001	1.8 (1.6, 2.0)	-1.6 (-3.3, 0.2)	.07	2.6 (2.0, 3.4)	- ^g		2.1 (2.0, 2.2)	-1.6 (-2.2, -1.1) ^f	<.001
AYA's (ages 15-39 y)		8.9 (8.9, 9.0)	-1.0 (-1.2, -0.7) ^f	<.001	8.9 (8.8, 9.0)	-1.2 (-1.5, -0.9) ^f	<.001	11.4 (11.2, 11.7)	-1.6 (-1.8, -1.3) ^f	<.001	6.9 (6.7, 7.2)	-0.8 (-1.3, -0.4) ^f	.002	11.0 (9.8, 12.4)	-0.5 (-2.2, 1.1)	.50	8.1 (7.9, 8.2)	1.1 (0.6, 1.6) ^f	.001
Males																			
Lung and bronchus	1	49.3 (49.1, 49.4)	-4.8 (-5.4, -4.2) ^f	<.001	51.8 (51.7, 52.0)	-4.4 (-4.7, -4.1) ^f	<.001	60.4 (59.8, 61.0)	-5.7 (-6.3, -5.2) ^f	<.001	29.3 (28.7, 29.9)	-3.0 (-3.4, -2.5) ^f	<.001	46.5 (43.9, 49.2)	-1.5 (-2.4, -0.7) ^f	.002	24.1 (23.7, 24.5)	-5.4 (-8.0, -2.6) ^f	<.001
Prostate	2	19.1 (19.0, 19.2)	-0.3 (-1.6, 1.0)	.60	18.0 (17.9, 18.1)	0.1 (-1.2, 1.4)	.88	38.7 (38.2, 39.2)	-3.9 (-4.2, -3.6) ^f	<.001	8.6 (8.3, 9.0)	-2.3 (-2.8, -1.8) ^f	<.001	21.2 (19.3, 23.3)	-1.2 (-2.4, -0.1) ^f	.04	15.7 (15.4, 16.0)	-0.7 (-2.2, 0.7)	.32
Colon and rectum	3	16.6 (16.5, 16.7)	-2.3 (-2.5, -2.1) ^f	<.001	16.3 (16.2, 16.4)	-1.8 (-2.3, -1.2) ^f	<.001	23.8 (23.5, 24.2)	-2.6 (-2.8, -2.4) ^f	<.001	11.5 (11.1, 11.9)	-2.1 (-2.5, -1.7) ^f	<.001	23.1 (21.3, 25.0)	0.2 (-1.0, 1.4)	.77	14.1 (13.9, 14.4)	-1.7 (-1.9, -1.5) ^f	<.001
Pancreas	4	12.7 (12.6, 12.7)	0.3 (0.2, 0.4) ^f	<.001	13 (12.9, 13.1)	0.5 (0.4, 0.6) ^f	<.001	15.3 (15.0, 15.6)	-0.2 (-0.4, 0.0)	.06	8.2 (7.9, 8.5)	0.0 (-0.5, 0.4)	.98	11.9 (10.6, 13.3)	0.0 (-1.6, 1.7)	.95	9.3 (9.1, 9.5)	-0.1 (-0.5, 0.4)	.69
Liver and intrahepatic bile duct	5	9.6 (9.6, 9.7)	0.5 (-0.5, 1.5)	.28	8.4 (8.3, 8.4)	1.0 (0.4, 1.6) ^f	.004	13.5 (13.3, 13.8)	-0.3 (-1.4, 0.9)	.65	13.5 (13.1, 13.9)	-3.7 (-4.9, -2.4) ^f	<.001	17 (15.5, 18.5)	2.8 (1.7, 4.0) ^f	<.001	13.2 (12.9, 13.5)	1.2 (0.9, 1.6) ^f	<.001
Leukemia	6	8.6 (8.5, 8.6)	-2.7 (-3.5, -2.0) ^f	<.001	9.1 (9.0, 9.2)	-2.2 (-2.8, -1.6) ^f	<.001	7.2 (7.0, 7.4)	-1.8 (-2.2, -1.3) ^f	<.001	4.8 (4.5, 5.0)	-0.6 (-1.4, 0.3)	.17	6.0 (5.1, 7.0)	-0.5 (-2.4, 1.4)	.56	5.8 (5.6, 6.0)	-1.0 (-1.5, -0.5) ^f	.001
Urinary bladder	7	7.5 (7.4, 7.6)	-1.1 (-2.0, -0.2) ^f	.02	8.3 (8.2, 8.4)	-0.8 (-1.9, 0.2)	.11	5.5 (5.3, 5.6)	-0.3 (-0.8, 0.2)	.27	2.8 (2.7, 3.0)	-0.4 (-1.3, 0.6)	.43	4.6 (3.8, 5.6)	- ^g		3.9 (3.7, 4.1)	-0.7 (-1.4, 0.0)	.06
Non-Hodgkin lymphoma	8	7.1 (7.1, 7.2)	-2.0 (-2.1, -1.8) ^f	<.001	7.5 (7.4, 7.6)	-1.9 (-2.2, -1.7) ^f	<.001	5.3 (5.1, 5.5)	-1.8 (-2.3, -1.2) ^f	<.001	4.9 (4.7, 5.2)	-1.4 (-1.9, -0.8) ^f	<.001	6.3 (5.3, 7.3)	0.2 (-1.7, 2.1)	.84	5.9 (5.7, 6.1)	-1.4 (-1.8, -1.0) ^f	<.001
Esophagus	9	7.0 (6.9, 7.1)	-1.2 (-1.4, -1.0) ^f	<.001	7.8 (7.8, 7.9)	-0.5 (-0.7, -0.3) ^f	<.001	5.5 (5.3, 5.6)	-4.8 (-5.2, -4.5) ^f	<.001	2.8 (2.6, 3.0)	-1.4 (-2.3, -0.5) ^f	.006	7.1 (6.1, 8.2)	-0.7 (-2.6, 1.2)	.42	3.6 (3.5, 3.8)	-1.4 (-2.1, -0.6) ^f	.001
Kidney and renal pelvis	10	5.4 (5.4, 5.5)	-2.4 (-3.0, -1.8) ^f	<.001	5.6 (5.5, 5.7)	-2.3 (-4.2, -0.5) ^f	.02	5.6 (5.4, 5.8)	-0.9 (-1.3, -0.5) ^f	<.001	2.6 (2.4, 2.8)	-0.1 (-1.3, 1.1)	.85	9.7 (8.5, 10.9)	-0.7 (-2.1, 0.7)	.28	5.0 (4.8, 5.1)	-0.8 (-1.4, -0.3) ^f	.007
Brain and other nervous system	11	5.4 (5.3, 5.4)	0.4 (0.1, 0.8) ^f	.01	6.1 (6.1, 6.2)	0.7 (0.4, 1.0) ^f	<.001	3.3 (3.2, 3.4)	0.3 (-0.4, 0.9)	.40	2.6 (2.5, 2.8)	0.3 (-0.6, 1.2)	.48	3.2 (2.6, 4.0)	0.7 (-1.7, 3.2)	.53	3.5 (3.4, 3.6)	0.4 (-0.1, 0.9)	.13
Myeloma	12	4.1 (4.1, 4.2)	-0.9 (-1.1, -0.7) ^f	<.001	3.9 (3.9, 4.0)	-0.9 (-1.1, -0.6) ^f	<.001	7.7 (7.4, 7.9)	-0.9 (-1.2, -0.6) ^f	<.001	2.0 (1.9, 2.2)	0.1 (-0.9, 1.1)	.81	4.1 (3.3, 5.0)	-1.6 (-3.3, 0.1)	.06	3.4 (3.2, 3.5)	-0.4 (-1.1, 0.3)	.28
Stomach	13	4.1 (4.1, 4.1)	-2.5 (-2.7, -2.2) ^f	<.001	3.2 (3.1, 3.2)	-2.7 (-3.1, -2.2) ^f	<.001	8.0 (7.8, 8.3)	-3.1 (-3.4, -2.8) ^f	<.001	6.7 (6.4, 7.0)	-3.6 (-4.0, -3.1) ^f	<.001	7.6 (6.5, 8.7)	-2.9 (-4.8, -1.0) ^f	.005	6.4 (6.2, 6.6)	-2.8 (-3.2, -2.4) ^f	<.001
Oral cavity and pharynx	14	3.9 (3.9, 4.0)	1.0 (0.5, 1.5) ^f	.002	4.0 (4.0, 4.1)	1.4 (0.9, 1.9) ^f	<.001	4.6 (4.5, 4.8)	-2.9 (-3.5, -2.2) ^f	<.001	3.1 (3.0, 3.3)	1.3 (-0.3, 2.9)	.10	4.0 (3.3, 4.8)	0.0 (-2.4, 2.4)	.98	2.4 (2.3, 2.5)	1.2 (-0.7, 3.0)	.20
Melanoma of the skin	15	3.5 (3.5, 3.6)	-6.1 (-7.6, -4.5) ^f	<.001	4.5 (4.4, 4.5)	-5.7 (-7.2, -4.1) ^f	<.001	0.4 (0.4, 0.5)	-1.4 (-3.4, 0.6)	.15	0.4 (0.3, 0.4)	- ^g		1.2 (0.8, 1.7)	- ^g		0.9 (0.8, 1.0)	-3.4 (-5.7, -1.1) ^f	.009
Nonmelanoma skin	16	1.7 (1.7, 1.8)	0.7 (-3.1, 4.6)	.73	2.0 (2.0, 2.0)	1.5 (-2.3, 5.4)	.44	0.7 (0.6, 0.8)	-2.7 (-3.8, -1.6) ^f	<.001	0.4 (0.3, 0.5)	- ^g		1.0 (0.7, 1.5)	- ^g		0.8 (0.7, 0.9)	0.6 (-0.7, 2.0)	.34
Larynx	17	1.7 (1.7, 1.8)	-2.5 (-2.7, -2.3) ^f	<.001	1.7 (1.6, 1.7)	-2.1 (-2.3, -1.9) ^f	<.001	3.1 (3.0, 3.3)	-3.6 (-4.0, -3.2) ^f	<.001	0.6 (0.5, 0.7)	-2.9 (-4.8, -0.9) ^f	.007	1.9 (1.4, 2.5)	- ^g		1.4 (1.3, 1.5)	-4.4 (-6.5, -2.3) ^f	.002

TABLE 3. Continued

Sex/Cancer Site or Type	Rank	All Races			Non-Hispanic White			Non-Hispanic Black			Non-Hispanic API			Non-Hispanic AI/AN (PRODA)			Hispanic		
		Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P
Soft tissue, including heart	18	1.5 (1.5, 1.6)	0.5 (0.3, 0.8) ^f	.001	1.6 (1.6, 1.6)	0.6 (0.2, 0.9) ^f	.002	1.6 (1.5, 1.7)	0.5 (0.4, 1.4)	.24	1.0 (0.9, 1.1)	0.5 (−1.4, 2.4)	.58	1.7 (1.2, 2.2)	− ^g	.04	1.2 (1.1, 1.3)	1.1 (0.0, 2.1) ^f	.04
Bones and Joints	19	0.6 (0.5, 0.6)	0.4 (−0.1, 0.8)	.08	0.6 (0.6, 0.6)	0.4 (−0.1, 0.9)	.13	0.6 (0.5, 0.7)	0.5 (0.5, 1.5)	.29	0.3 (0.3, 0.4)	2.4 (0.2, 4.8) ^f	.04	0.7 (0.4, 1.0)	− ^g	.97	0.5 (0.4, 0.5)	0.0 (−1.5, 1.6)	.97
Females																			
Lung and bronchus	1	33.2 (33.1, 33.3)	−3.7 (−4.3, −3.1) ^f	<.001	36.8 (36.7, 36.9)	−3.3 (−3.9, −2.7) ^f	<.001	31.9 (31.6, 32.3)	−4.6 (−5.7, −3.4) ^f	<.001	16.9 (16.6, 17.3)	−3.7 (−5.9, −1.4) ^f	.002	32.6 (30.7, 34.5)	−2.2 (−3.2, −1.3) ^f	<.001	12.6 (12.4, 12.9)	−3.7 (−6.5, −0.9) ^f	.01
Breast	2	20.3 (20.2, 20.4)	−1.2 (−1.4, −1.0) ^f	<.001	20.3 (20.2, 20.4)	−1.1 (−1.4, −0.8) ^f	<.001	28.4 (28.1, 28.8)	−1.5 (−1.6, −1.4) ^f	<.001	11.5 (11.2, 11.8)	−0.7 (−1.2, −0.3) ^f	.003	16.6 (15.2, 17.9)	3.9 (−2.8, 11.0)	.23	14.0 (13.7, 14.2)	−1.1 (−1.4, −0.9) ^f	<.001
Colon and rectum	3	11.7 (11.7, 11.8)	−1.6 (−2.3, −0.8) ^f	.001	11.7 (11.7, 11.8)	−1.6 (−2.1, −1.1) ^f	<.001	15.6 (15.4, 15.9)	−3.1 (−3.3, −2.9) ^f	<.001	8.1 (7.9, 8.4)	−2.0 (−2.6, −1.5) ^f	<.001	15 (13.7, 16.3)	−0.6 (−2.0, 0.7)	.34	8.7 (8.5, 8.9)	−2.1 (−2.4, −1.9) ^f	<.001
Pancreas	4	9.6 (9.5, 9.6)	0.2 (0.1, 0.3) ^f	.001	9.6 (9.5, 9.6)	0.3 (0.2, 0.5) ^f	<.001	12.3 (12.1, 12.5)	−0.2 (−0.4, 0.0) ^f	.03	7.1 (6.8, 7.3)	0.1 (−0.3, 0.5)	.71	8.9 (7.9, 9.9)	0.3 (−1.4, 2.0)	.70	7.8 (7.6, 7.9)	0.0 (−0.2, 0.3)	.95
Ovary	5	6.9 (6.8, 6.9)	−2.3 (−2.5, −2.2) ^f	<.001	7.3 (7.2, 7.4)	−2.4 (−2.6, −2.2) ^f	<.001	6.1 (6.0, 6.3)	−1.6 (−2.0, −1.3) ^f	<.001	4.5 (4.3, 4.6)	−0.9 (−1.4, −0.4) ^f	.002	7.1 (6.3, 8.0)	−1.1 (−2.3, 0.2)	.10	5.1 (5.0, 5.3)	−1.5 (−1.8, −1.1) ^f	<.001
Corpus and uterus, NOS	6	4.8 (4.8, 4.9)	2.1 (1.7, 2.4) ^f	<.001	4.5 (4.4, 4.5)	1.9 (1.5, 2.4) ^f	<.001	8.8 (8.6, 9.0)	2.3 (1.8, 2.9) ^f	<.001	3.2 (3.0, 3.3)	2.6 (1.9, 3.4) ^f	<.001	4.0 (3.4, 4.7)	− ^g	<.001	4.0 (3.9, 4.1)	2.6 (2.0, 3.3) ^f	<.001
Leukemia	7	4.8 (4.8, 4.8)	−1.3 (−1.5, −1.2) ^f	<.001	5.0 (5.0, 5.1)	−1.2 (−1.4, −1.1) ^f	<.001	4.4 (4.3, 4.6)	−1.5 (−1.8, −1.1) ^f	<.001	2.6 (2.5, 2.8)	−4.6 (−7.4, −1.6) ^f	.006	3.5 (2.9, 4.1)	−2.1 (−4.4, 0.3)	.08	3.7 (3.6, 3.9)	−0.8 (−1.2, −0.3) ^f	.002
Non-Hodgkin lymphoma	8	4.2 (4.2, 4.3)	−2.6 (−2.8, −2.5) ^f	<.001	4.5 (4.4, 4.5)	−2.2 (−2.8, −1.6) ^f	<.001	3.3 (3.2, 3.4)	−2.2 (−2.5, −1.8) ^f	<.001	3.0 (2.8, 3.1)	−2.1 (−2.7, −1.4) ^f	<.001	3.7 (3.1, 4.4)	−3.4 (−5.3, −1.5) ^f	.002	3.7 (3.6, 3.8)	−2.0 (−2.6, −1.5) ^f	<.001
Liver and Intrahepatic bile duct	9	4.0 (3.9, 4.0)	1.6 (1.0, 2.2) ^f	.001	3.5 (3.5, 3.5)	2.2 (2.0, 2.4) ^f	<.001	4.9 (4.7, 5.0)	1.8 (1.3, 2.3) ^f	<.001	5.6 (5.4, 5.8)	−1.3 (−2.1, −0.6) ^f	.001	8.1 (7.2, 9.1)	1.0 (−1.1, 3.1)	.32	6.0 (5.8, 6.1)	1.2 (0.8, 1.5) ^f	<.001
Brain and other nervous system	10	3.6 (3.6, 3.6)	0.5 (0.2, 0.8) ^f	.003	4.1 (4.1, 4.2)	0.7 (0.3, 1.1) ^f	.002	2.2 (2.1, 2.3)	0.4 (−0.3, 1.1)	.21	1.9 (1.8, 2.1)	2.1 (1.0, 3.2) ^f	.001	2.7 (2.2, 3.3)	− ^g	.02	2.6 (2.5, 2.7)	0.6 (0.1, 1.1) ^f	.02
Myeloma	11	2.6 (2.6, 2.6)	−1.9 (−3.0, −0.9) ^f	.003	2.3 (2.3, 2.4)	−1.7 (−2.7, −0.7) ^f	.004	5.4 (5.3, 5.6)	−1.0 (−1.6, −0.5) ^f	.001	1.2 (1.1, 1.3)	−1.2 (−2.7, 0.3)	.10	3.3 (2.7, 4.0)	−1.1 (−3.9, 1.7)	.41	2.2 (2.1, 2.3)	−1.4 (−2.0, −0.8) ^f	<.001
Kidney and renal pelvis	12	2.3 (2.3, 2.3)	−1.4 (−1.6, −1.3) ^f	<.001	2.4 (2.3, 2.4)	−1.4 (−1.6, −1.2) ^f	<.001	2.3 (2.2, 2.4)	−1.6 (−2.0, −1.1) ^f	<.001	1.1 (1.0, 1.2)	−1.0 (−2.1, 0.1)	.06	4.0 (3.3, 4.7)	−1.6 (−3.5, 0.3)	.09	2.2 (2.1, 2.3)	−0.7 (−1.2, −0.1) ^f	.03
Cervix uteri	13	2.3 (2.2, 2.3)	−0.7 (−0.9, −0.4) ^f	<.001	2.1 (2.0, 2.1)	−0.1 (−0.4, 0.2)	.43	3.6 (3.4, 3.7)	−2.4 (−2.8, −2.1) ^f	<.001	1.7 (1.6, 1.9)	−2.2 (−3.2, −1.1) ^f	<.001	3.1 (2.6, 3.7)	−1.7 (−3.6, 0.2)	.08	2.6 (2.5, 2.7)	−0.2 (−1.8, 1.4)	.75
Stomach	14	2.2 (2.2, 2.2)	−1.8 (−2.2, −1.4) ^f	<.001	1.6 (1.6, 1.6)	−3.0 (−3.2, −2.7) ^f	<.001	3.7 (3.6, 3.8)	−3.4 (−3.8, −3.1) ^f	<.001	4.1 (3.9, 4.3)	−3.1 (−4.6, −1.5) ^f	<.001	4.1 (3.4, 4.8)	−2.9 (−4.7, −1.1) ^f	.005	4.0 (3.8, 4.1)	−1.9 (−2.4, −1.4) ^f	<.001
Urinary bladder	15	2.1 (2.1, 2.2)	−0.5 (−0.7, −0.4) ^f	<.001	2.3 (2.2, 2.3)	−0.2 (−0.4, 0.0) ^f	.02	2.4 (2.3, 2.5)	−1.4 (−1.9, −1.0) ^f	<.001	0.9 (0.8, 1.0)	−1.0 (−2.1, 0.0)	.06	1.9 (1.4, 2.4)	− ^g	.08	1.3 (1.2, 1.3)	−0.7 (−1.6, 0.1)	.08
Melanoma of the skin	16	1.5 (1.4, 1.5)	−6.3 (−8.6, −3.8) ^f	<.001	1.9 (1.9, 1.9)	−5.9 (−8.5, −3.2) ^f	.001	0.3 (0.3, 0.3)	−2.2 (−3.7, −0.7) ^f	.009	0.3 (0.2, 0.3)	−0.4 (−3.3, 2.6)	.77	0.5 (0.3, 0.8)	− ^g	.02	0.5 (0.5, 0.6)	−1.6 (−2.9, −0.4) ^f	.02
Esophagus	17	1.4 (1.4, 1.5)	−1.6 (−1.8, −1.4) ^f	<.001	1.5 (1.5, 1.5)	−0.9 (−1.1, −0.7) ^f	<.001	1.7 (1.6, 1.8)	−4.3 (−4.8, −3.8) ^f	<.001	0.6 (0.6, 0.7)	−2.3 (−3.9, −0.7) ^f	.008	1.4 (1.1, 1.9)	− ^g	<.001	0.7 (0.7, 0.8)	−2.2 (−3.0, −1.4) ^f	<.001
Oral cavity and pharynx	18	1.3 (1.3, 1.4)	1.4 (−1.4, 4.3)	.31	1.4 (1.4, 1.4)	2.0 (−0.5, 4.6)	.10	1.3 (1.3, 1.4)	−2.1 (−2.8, −1.4) ^f	<.001	1.1 (1.0, 1.2)	−1.5 (−2.9, −0.2) ^f	.03	1.2 (0.9, 1.6)	− ^g	.71	0.8 (0.7, 0.9)	−0.2 (−1.2, 0.8)	.71
Soft tissue, including heart	19	1.2 (1.1, 1.2)	0.2 (0.0, 0.3) ^f	.02	1.2 (1.1, 1.2)	0.1 (−0.1, 0.3)	.45	1.5 (1.4, 1.6)	0.8 (0.5, 1.2) ^f	<.001	0.8 (0.7, 0.9)	0.8 (−0.7, 2.4)	.26	1.1 (0.8, 1.5)	− ^g	.27	0.9 (0.9, 1.0)	0.6 (−0.5, 1.6)	.27
Gallbladder	20	0.7 (0.7, 0.7)	−1.2 (−1.5, −0.8) ^f	<.001	0.6 (0.6, 0.6)	−1.7 (−2.1, −1.4) ^f	<.001	1.0 (1.0, 1.1)	0.0 (−0.8, 0.8)	.94	0.8 (0.7, 0.8)	−0.9 (−2.1, 0.4)	.17	1.8 (1.4, 2.3)	−2.6 (−4.6, −0.5) ^f	.02	1.1 (1.1, 1.2)	−1.6 (−2.5, −0.6) ^f	.003

TABLE 3. Continued

Sex/Cancer Site or Type	All Races			Non-Hispanic White			Non-Hispanic Black			Non-Hispanic API			Non-Hispanic AI/AN (PRODA)			Hispanic		
	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P	Rate (95% CI)	AAPC (95% CI)	P
Children																		
Brain and other nervous system	0.7 (0.7, 0.7)	-0.2 (-0.7, 0.2)	.27	0.7 (0.7, 0.8)	-0.4 (-1.0, 0.2)	.17	0.7 (0.7, 0.8)	0.5 (-0.9, 1.9)	.48	0.5 (0.4, 0.7)	-0.9 (-3.2, 1.4)	.41	0.9 (0.6, 1.4)	- ^g		0.7 (0.6, 0.7)	0.2 (-0.6, 0.9)	.66
Leukemia	0.5 (0.5, 0.6)	-2.8 (-3.5, -2.2) ^f	< .001	0.5 (0.5, 0.5)	-3.0 (-3.8, -2.3) ^f	< .001	0.5 (0.4, 0.6)	-2.4 (-3.7, -1.2) ^f	.001	0.5 (0.4, 0.6)	-3.7 (-5.6, -1.7) ^f	.001	0.8 (0.5, 1.2)	- ^g		0.7 (0.6, 0.7)	-3.3 (-4.4, -2.2) ^f	< .001
AYA ^s																		
Female breast	2.2 (2.1, 2.3)	0.6 (-0.8, 2.1)	.36	2.0 (2.0, 2.1)	0.9 (-1.3, 3.0)	.40	3.9 (3.7, 4.2)	-2.3 (-2.7, -2.0) ^f	< .001	1.3 (1.1, 1.5)	-0.7 (-2.7, 1.4)	.49	1.6 (1.0, 2.4)	- ^g		1.8 (1.7, 1.9)	-0.8 (-2.1, 0.5)	.22
Brain and other nervous system	1.0 (0.9, 1.0)	-0.2 (-0.7, 0.3)	.36	1.2 (1.2, 1.2)	-0.1 (-0.6, 0.5)	.78	0.6 (0.6, 0.7)	0.4 (-0.8, 1.6)	.51	0.6 (0.6, 0.7)	1.4 (-0.5, 3.3)	.14	0.8 (0.5, 1.2)	- ^g		0.6 (0.6, 0.7)	0.8 (-0.2, 1.8)	.12
Leukemia	0.9 (0.9, 1.0)	-2.2 (-2.6, -1.8) ^f	< .001	0.8 (0.8, 0.8)	-2.9 (-3.5, -2.4) ^f	< .001	1.0 (0.9, 1.1)	-2.4 (-3.2, -1.5) ^f	< .001	0.8 (0.7, 0.9)	-1.2 (-2.6, 0.3)	.10	1.1 (0.8, 1.5)	- ^g		1.3 (1.2, 1.3)	-1.1 (-1.7, -0.5) ^f	.002
Colon and rectum	0.9 (0.8, 0.9)	0.9 (0.5, 1.3) ^f	< .001	0.9 (0.9, 0.9)	1.5 (0.8, 2.2) ^f	< .001	1.2 (1.1, 1.3)	0.7 (-0.2, 1.5)	.11	0.7 (0.6, 0.8)	-0.3 (-2.2, 1.7)	.78	1.4 (1.0, 1.9)	- ^g		0.6 (0.6, 0.7)	1.3 (0.3, 2.3) ^f	.01

Abbreviations: AAPC, average annual percent change; AI/AN, American Indian/Alaska Native; API, Asian/Pacific Islander; AYAs, adolescents and young adults; CI, confidence interval; NOS, not otherwise specified; PRODA, Indian Health Service Purchased/Referred Care Delivery Area.

^aRates are per 100,000 persons and age standardized to the 2000 US standard population (19 age groups; US Bureau of the Census, Current Population Reports, Publication 25-1130, US Government Printing Office; 2000 [Census P25-1130]).

^bFixed interval trends are characterized as the AAPC, the weighted average of the APCs over the fixed interval 2013 to 2017, using the underlying Joinpoint model (with up to 3 joinpoints) for the period 2001 to 2017. The Joinpoint Regression Program was used (version 4.7.0.0; Statistical Research and Applications Branch, National Cancer Institute, February 2019).

^cCancers are listed in descending rank order of sex-specific, age-adjusted death rates for 2013 to 2017 for all racial/ethnic groups combined. More than 15 cancers may appear for males and females to include the top 15 cancers in each racial/ethnic group.

^dRacial/ethnic groups are mutually exclusive. Data for non-Hispanic AI/ANs are restricted to PRODA counties.

^eBased on data from the National Center for Health Statistics public-use data file for the total United States, 2001 to 2017.

^fThe AAPC is statistically significantly different from zero ($P < .05$). Two-sided statistical significance for the AAPC was determined using a t test when the AAPC lay entirely within the last joinpoint segment or a z test when the AAPC extended beyond the last joinpoint segment.

^gThe statistic could not be calculated. The AAPC is based on <10 cases for at least 1 year within the time interval.

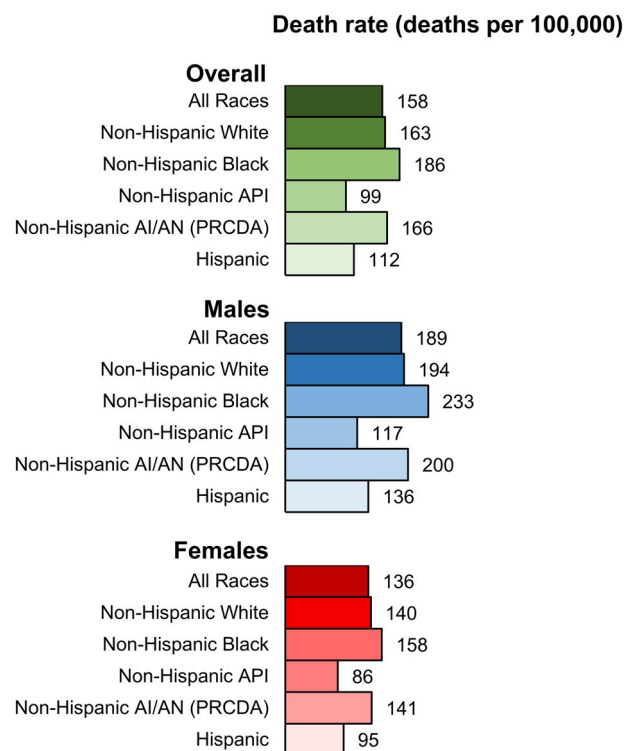


Figure 5. Age-standardized, overall cancer death rates for 2013 through 2017 are illustrated among males and females by racial/ethnic group. Racial/ethnic groups are mutually exclusive. Data for the non-Hispanic American Indian/Alaska Native (AI/AN) population are restricted to Indian Health Service Purchased/Referred Care Delivery Area (PRCDA) counties. API indicates Asian/Pacific Islander.

during 2001 through 2010 and then stabilized during 2010 through 2017.

DISCUSSION

Although overall cancer death rates continued to decline among both males and females, incidence rates leveled off among males and increased slightly among females. These trends reflect population changes in cancer risk factors, screening test use, diagnostic practices, and treatment advances.² Part II of this report showed that Healthy People 2020 objectives were not yet met overall for major cancer risk factors such as excess body weight, excessive alcohol use, and cigarette smoking.² Historical declines in cigarette smoking have been reflected by declines in incidence of and mortality from several tobacco-related cancers, including lung, larynx, and bladder, which have greatly affected the overall incidence and death rates.^{14,15} Although the decrease in lung cancer was substantial, it continues to be the leading cause of cancer death, accounting for about

one-quarter of all cancer deaths. Furthermore, these gains are being offset by increasing incidence trends for cancers related to excess body weight and physical inactivity, including those for cancers of the female breast, uterus, kidney, liver, and pancreas.^{16,17}

Through early detection of cancer and subsequent timely and effective treatment, the use of cancer screening tests can reduce mortality.¹⁸⁻²³ However, the effects on incidence are more complex because screening can lead to either increases or decreases in the number of diagnosed cases.²⁴ For example, the number of cases may increase through the detection of indolent, slow-growing, or inconsequential tumors, or the number of malignant colorectal and cervical cases may decrease when screening tests detect precancerous lesions that are treated before progressing to cancer. Part II of this report shows that the current use of recommended screening tests is suboptimal, ranging from <10% for lung cancer to 60% for colorectal cancer, 72% for breast cancer, and 81% for cervical cancer.^{2,25} Furthermore, breast and cervical cancer screening use has declined.^{2,25} This report shows that rates of cervical cancer incidence have leveled off after over a decade of sustained declines, which may coincide with declines in screening rates.

For the first time, the Annual Report to the Nation provided rates and trends for the most common cancers among children and AYAs. As noted in last year's report, cancers among these age groups are of particular significance because many young patients experience long-term sequelae of their cancer and its treatment, affecting their health and quality of life for the remainder of their lifetimes.¹ Reasons for increasing incidence rates for cancer overall and for the most common types among children are largely unknown.²⁶ Factors potentially influencing childhood cancer incidence trends include changes over time in diagnostic technology, disease classification, and registry completeness²⁶ as well as changes in risk factors, such as increasing maternal age.²⁷ Continued declines in cancer death rates among children likely result from improved treatment. As part of an effort to provide data faster for pediatric cancer research and more quickly identify young patients eligible for clinical trial enrollment, the CDC created a cooperative program to help cancer registries improve rapid reporting of pediatric and young adult cases.²⁸ The NCI is launching an initiative to enhance data collection for childhood cancers.²⁹

Among AYAs, the increase in female breast cancer incidence rates during 2010 through 2016 after a period

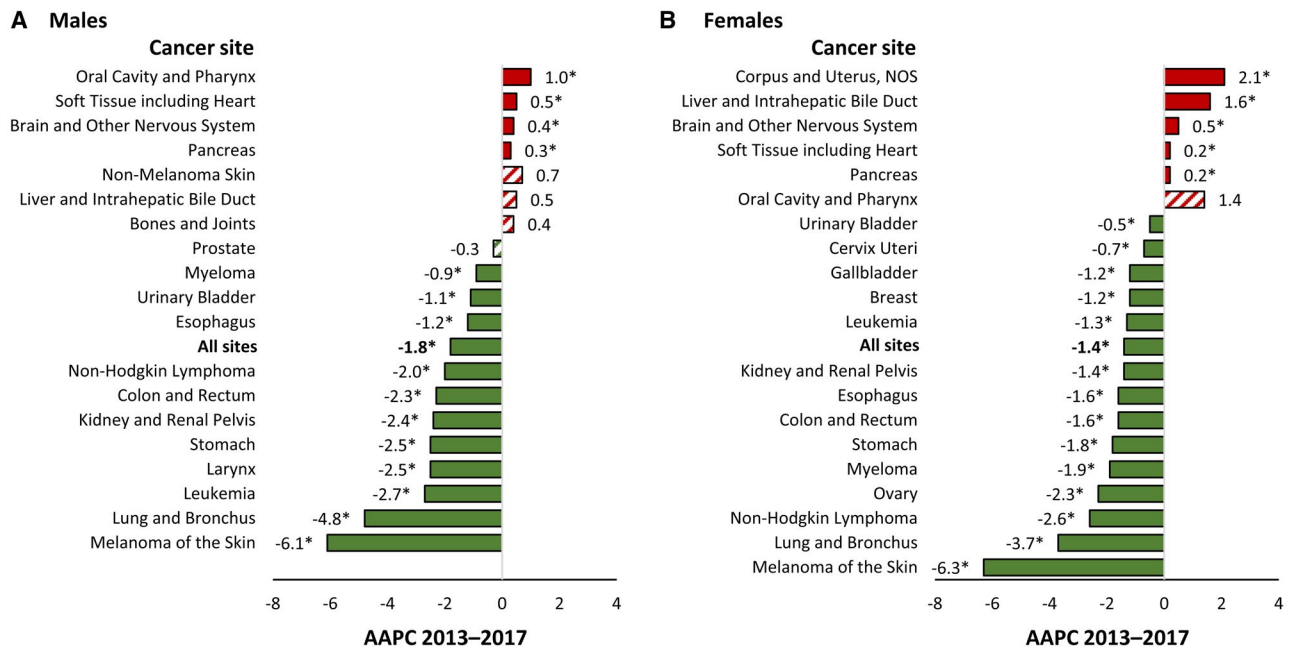


Figure 6. The average annual percent change (AAPC) in age-standardized death rates for 2013 through 2017 are illustrated for all sites and for the 19 most common cancer deaths in men and the 20 most common cancer deaths in women, all ages, all races/ethnicities combined, among (A) males and (B) females. The AAPC was a weighted average of the annual percent changes (APCs) over the fixed 5-year interval from 2013 to 2017 using the underlying joinpoint regression model, which allowed up to 4 different APCs, for the 17-year period from 2001 to 2017. AAPCs with an asterisk (*) were statistically significantly different from zero ($P < .05$) and are depicted as solid-colored bars; AAPCs in bars with hash marks were not statistically significantly different from zero (stable). NOS indicates not otherwise specified.

of stability is concerning—particularly because death rates, which had declined during 2001 through 2010, also began to stabilize in 2010. As noted in last year's report, opportunities for primary prevention of breast cancer among younger and premenopausal women are limited.¹ More detailed studies may help evaluate potential reasons for these adverse trends.

During the most recent 5-year period in this report, colorectal cancer death rates declined among both males and females overall, most likely caused in part by increases in the use of guideline-concordant colorectal cancer screening,³⁰ which rose from 39% in 2000 to 55% in 2008 and continued to increase, but at a slower pace, to 61% in 2015.³¹ The relatively rapid declines in colorectal cancer incidence rates during 2001 through 2012 observed in this report correspond with this timeline of increased use of colorectal cancer screening. However, during 2012 through 2016, colorectal cancer incidence rates stabilized among women, and the decline in rates slowed among men, consistent with the slowing pace in colorectal cancer screening use and increasing prevalence of risk factors such as excess body weight, adult weight gain, and diabetes.³⁰ This year's report observed a

continued increase in colorectal cancer death rates among AYAs, which has been noted previously.^{32,33}

Excess body weight may have contributed to increasing trends in uterine cancer incidence and death rates.³⁴ This report shows that, as death rates from leukemia and non-Hodgkin lymphoma continue to decrease, uterine cancer is now ranked as the sixth most common cause of cancer death among women (slightly higher than leukemia). Excess body weight is also a risk factor for liver cancer.^{35,36} Liver cancer death rates continued to increase among females overall but stabilized among males, reflecting differing trends by racial/ethnic group. Liver cancer death rates have decreased among APIs largely because of improvements in hepatitis B virus vaccination coverage and treatment.³⁷ However, prevalent risk factors for liver cancer, including excess body weight, diabetes, nonalcoholic fatty liver disease, alcohol use, and hepatitis C virus infection, may contribute to increases in liver cancer death rates observed in other groups.³⁷ Incident rates for cancers of the oral cavity and pharynx increased among non-Hispanic white men and women and API men. Because tobacco use, a prominent risk factor, has declined, researchers suggest that the increase in this

TABLE 4. Joinpoint Trends^a for the Most Common Causes of Cancer Death,^b All Races/Ethnicities Combined by Sex and Age Group, United States,^c 2001 to 2017

Sex/Cancer Site or Type	Years	Trend 1			Trend 2			Trend 3			Trend 4			Fixed Interval 2013-2017	
		APC (95% CI)	P	Years	APC (95% CI)	P	Years	APC (95% CI)	P	Years	APC (95% CI)	P	Years	AAPC (95% CI) ^d	P
All sites															
Both sexes	2001-2017	-1.5 (-1.6, -1.5) ^e	< .001											-1.5 (-1.6, -1.5) ^e	< .001
Males	2001-2017	-1.8 (-1.9, -1.8) ^e	< .001											-1.8 (-1.9, -1.8) ^e	< .001
Females	2001-2017	-1.4 (-1.4, -1.4) ^e	< .001											-1.4 (-1.4, -1.4) ^e	< .001
Children (ages 0-14 years)	2001-2017	-1.4 (-1.8, -1.1) ^e	< .001											-1.4 (-1.8, -1.1) ^e	< .001
AYAs (ages 15-39 years)	2001-2005	-3.0 (-4.0, -1.9) ^e	< .001	2005-2017	-1.0 (-1.2, -0.7) ^e	< .001								-1.0 (-1.2, -0.7) ^e	< .001
Males															
Lung and bronchus	2001-2005	-2.0 (-2.3, -1.7) ^e	< .001	2005-2012	-2.9 (-3.1, -2.8) ^e	< .001	2012-2015	-4.0 (-5.1, -3.0) ^e	< .001	2015-2017	-5.5 (-6.6, -4.5) ^e	< .001		-4.8 (-5.4, -4.2) ^e	< .001
Prostate	2001-2013	-3.5 (-3.7, -3.2) ^e	< .001	2013-2017	-0.3 (-1.6, 1.0)	.60								-0.3 (-1.6, 1.0)	.60
Colon and rectum	2001-2006	-3.6 (-4.2, -2.9) ^e	< .001	2006-2017	-2.3 (-2.5, -2.1) ^e	< .001								-2.3 (-2.5, -2.1) ^e	< .001
Pancreas	2001-2017	0.3 (0.2, 0.4) ^e	< .001											0.3 (0.2, 0.4) ^e	< .001
Liver and Intrahepatic bile duct	2001-2013	2.7 (2.5, 2.9) ^e	< .001	2013-2017	0.5 (-0.5, 1.5)	.28								0.5 (-0.5, 1.5)	.28
Leukemia	2001-2013	-1.0 (-1.1, -0.8) ^e	< .001	2013-2017	-2.7 (-3.5, -2.0) ^e	< .001								-2.7 (-3.5, -2.0) ^e	< .001
Urinary bladder	2001-2013	0.0 (-0.2, 0.2)	.73	2013-2017	-1.1 (-2.0, -0.2) ^e	.02								-1.1 (-2.0, -0.2) ^e	.02
Non-Hodgkin lymphoma	2001-2006	-3.0 (-3.7, -2.4) ^e	< .001	2006-2017	-2.0 (-2.1, -1.8) ^e	< .001								-2.0 (-2.1, -1.8) ^e	< .001
Esophagus	2001-2006	0.0 (-0.7, 0.7)	.96	2006-2017	-1.2 (-1.4, -1.0) ^e	< .001								-1.2 (-1.4, -1.0) ^e	< .001
Kidney and renal pelvis	2001-2006	-1.4 (-1.7, -1.0) ^e	< .001	2006-2015	-0.5 (-0.7, -0.3) ^e	< .001	2015-2017	-4.3 (-5.7, -3.0) ^e	< .001					-2.4 (-3.0, -1.8) ^e	< .001
Brain and other nervous system	2001-2006	-1.3 (-2.4, -0.1) ^e	.03	2006-2017	0.4 (0.1, 0.8) ^e	.01								0.4 (0.1, 0.8) ^e	.01
Myeloma	2001-2017	-0.9 (-1.1, -0.7) ^e	< .001											-0.9 (-1.1, -0.7) ^e	< .001
Stomach	2001-2006	-3.7 (-4.5, -2.9) ^e	< .001	2006-2017	-2.5 (-2.7, -2.2) ^e	< .001								-2.5 (-2.7, -2.2) ^e	< .001
Oral cavity and pharynx	2001-2009	-1.4 (-2.0, -0.8) ^e	< .001	2009-2017	1.0 (0.5, 1.5) ^e	.002								1.0 (0.5, 1.5) ^e	.002
Melanoma of the skin	2001-2009	1.0 (0.5, 1.6) ^e	.002	2009-2014	-1.4 (-2.8, 0.0)	.06	2014-2017	-7.6 (-9.8, -5.3) ^e	< .001					-6.1 (-7.6, -4.5) ^e	< .001

TABLE 4. Continued

Sex/Cancer Site or Type	Years	Trend 1			Trend 2			Trend 3			Trend 4			Fixed Interval 2013-2017	
		APC (95% CI)	P	Years	APC (95% CI)	P	Years	APC (95% CI)	P	Years	APC (95% CI)	P	Years	AAPC (95% CI) ^d	P
Nonmelanoma skin	2001-2004	-1.4 (-5.6, 3.0)	.46	2004-2012	1.6 (0.5, 2.7) ^e	.01	2012-2015	4.2 (-2.8, 11.7)	.20	2015-2017	-2.8 (-9.0, 3.9)	.34		0.7 (-3.1, 4.6)	.73
Larynx	2001-2017	-2.5 (-2.7, -2.3) ^e	<.001											-2.5 (-2.7, -2.3) ^e	<.001
Soft tissue, including heart	2001-2017	0.5 (0.3, 0.8) ^e	.001											0.5 (0.3, 0.8) ^e	.001
Bones and joints	2001-2017	0.4 (-0.1, 0.8)	.08											0.4 (-0.1, 0.8)	.08
Females															
Lung and bronchus	2001-2007	-0.5 (-0.9, -0.2) ^e	.004	2007-2014	-2.0 (-2.3, -1.7) ^e	<.001	2014-2017	-4.2 (-5.1, -3.3) ^e	<.001					-3.7 (-4.3, -3.1) ^e	<.001
Breast	2001-2003	-1.6 (-2.2, -0.9) ^e	.001	2003-2007	-2.3 (-2.6, -1.9) ^e	<.001	2007-2013	-1.6 (-1.8, -1.5) ^e	<.001	2013-2017	-1.2 (-1.4, -1.0) ^e	<.001		-1.2 (-1.4, -1.0) ^e	<.001
Colon and rectum	2001-2012	-2.9 (-3.1, -2.7) ^e	<.001	2012-2017	-1.6 (-2.3, -0.8) ^e	.001								-1.6 (-2.3, -0.8) ^e	.001
Pancreas	2001-2017	0.2 (0.1, 0.3) ^e	.001											0.2 (0.1, 0.3) ^e	.001
Ovary	2001-2005	-1.1 (-1.9, -0.2) ^e	.02	2005-2017	-2.3 (-2.5, -2.2) ^e	<.001								-2.3 (-2.5, -2.2) ^e	<.001
Corpus and uterus, NOS	2001-2008	0.1 (-0.4, 0.7)	.58	2008-2017	2.1 (1.7, 2.4) ^e	<.001								2.1 (1.7, 2.4) ^e	<.001
Leukemia	2001-2017	-1.3 (-1.5, -1.2) ^e	<.001											-1.3 (-1.5, -1.2) ^e	<.001
Non-Hodgkin lymphoma	2001-2005	-3.6 (-4.5, -2.8) ^e	<.001	2005-2017	-2.6 (-2.8, -2.5) ^e	<.001								-2.6 (-2.8, -2.5) ^e	<.001
Liver and intrahepatic bile duct	2001-2004	2.8 (1.0, 4.5) ^e	.008	2004-2008	0.5 (-1.1, 2.1)	.46	2008-2012	3.6 (2.1, 5.2) ^e	.001	2012-2017	1.6 (1.0, 2.2) ^e	.001		1.6 (1.0, 2.2) ^e	.001
Brain and other nervous system	2001-2006	-1.2 (-2.1, -0.2) ^e	.02	2006-2017	0.5 (0.2, 0.8) ^e	.003								0.5 (0.2, 0.8) ^e	.003
Myeloma	2001-2009	-2.5 (-3.0, -2.0) ^e	<.001	2009-2012	1.7 (-3.2, 6.8)	.46	2012-2017	-1.9 (-3.0, -0.9) ^e	.003					-1.9 (-3.0, -0.9) ^e	.003
Kidney and renal pelvis	2001-2017	-1.4 (-1.6, -1.3) ^e	<.001											-1.4 (-1.6, -1.3) ^e	<.001
Cervix uteri	2001-2004	-3.0 (-5.1, -0.8) ^e	.01	2004-2017	-0.7 (-0.9, -0.4) ^e	<.001								-0.7 (-0.9, -0.4) ^e	<.001
Stomach	2001-2008	-3.0 (-3.5, -2.5) ^e	<.001	2008-2017	-1.8 (-2.2, -1.4) ^e	<.001								-1.8 (-2.2, -1.4) ^e	<.001
Urinary bladder	2001-2017	-0.5 (-0.7, -0.4) ^e	<.001											-0.5 (-0.7, -0.4) ^e	<.001
Melanoma of the skin	2001-2013	-0.4 (-0.9, 0.1)	.08	2013-2017	-6.3 (-8.6, -3.8) ^e	<.001								-6.3 (-8.6, -3.8) ^e	<.001
Esophagus	2001-2017	-1.6 (-1.8, -1.4) ^e	<.001											-1.6 (-1.8, -1.4) ^e	<.001

TABLE 4. Continued

Sex/Cancer Site or Type	Years	Trend 1			Trend 2			Trend 3			Trend 4			Fixed Interval 2013–2017		
		APC (95% CI)	P	Years	APC (95% CI)	P	Years	APC (95% CI)	P	Years	APC (95% CI)	P	Years	AAPC (95% CI) ^d	P	Years
Oral cavity and pharynx	2001–2013	–1.4 (–1.9, –0.9) ^e	< .001	2013–2017	1.4 (–1.4, 4.3)	.31								1.4 (–1.4, 4.3)	.31	
Soft tissue, in- cluding heart	2001–2017	0.2 (0.0, 0.3) ^e	.02											0.2 (0.0, 0.3) ^e	.02	
Gallbladder	2001–2017	–1.2 (–1.5, –0.8) ^e	< .001											–1.2 (–1.5, –0.8) ^e	< .001	
Children																
Brain and other nervous system	2001–2017	–0.2 (–0.7, 0.2)	.27											–0.2 (–0.7, 0.2)	.27	
Leukemia	2001–2017	–2.8 (–3.5, –2.2) ^e	< .001											–2.8 (–3.5, –2.2) ^e	< .001	
AYAs																
Female breast	2001–2010	–3.2 (–4.1, –2.3) ^e	< .001	2010–2017	0.6 (–0.8, 2.1)	.36								0.6 (–0.8, 2.1)	.36	
Brain and other nervous system	2001–2017	–0.2 (–0.7, 0.3)	.36											–0.2 (–0.7, 0.3)	.36	
Leukemia	2001–2017	–2.2 (–2.6, –1.8) ^e	< .001											–2.2 (–2.6, –1.8) ^e	< .001	
Colon and rectum	2001–2017	0.9 (0.5, 1.3) ^e	< .001											0.9 (0.5, 1.3) ^e	< .001	

Abbreviations: AAPC, average annual percent change; APC, annual percent change; AYAs, adolescents and young adults; CI, confidence interval; NOS, not otherwise specified.

^aJoinpoint models with up to 3 joinpoints are based on rates per 100,000 persons, age standardized to the 2000 US standard population (19 age groups; US Bureau of the Census. Current Population Reports, Publication 25-1130. US Government Printing Office; 2000 [Census P25-1130]). The Joinpoint Regression Program was used (version 4.7.0.0; Statistical Research and Applications Branch, National Cancer Institute; February 2019).

^bCancers are listed in descending rank order of sex-specific, age-adjusted death rates for 2013 to 2017 for all racial/ethnic groups combined. More than 15 cancers may appear for males and females to include the top 15 cancers in each racial/ethnic group.

^cBased on data from the National Center for Health Statistics public-use data file for the total United States, 2001 to 2017.

^dThe AAPC is the weighted average of the APCs over the fixed interval 2013 to 2017 using the underlying Joinpoint model for the period 2001 to 2017.

^eThe APC or AAPC is statistically significantly different from zero ($P < .05$). Two-sided statistical significance for the APC was determined using a t test and, for the AAPC, using a t test when the AAPC lay entirely within the last joinpoint segment or a z test when the AAPC extended beyond the last joinpoint segment.

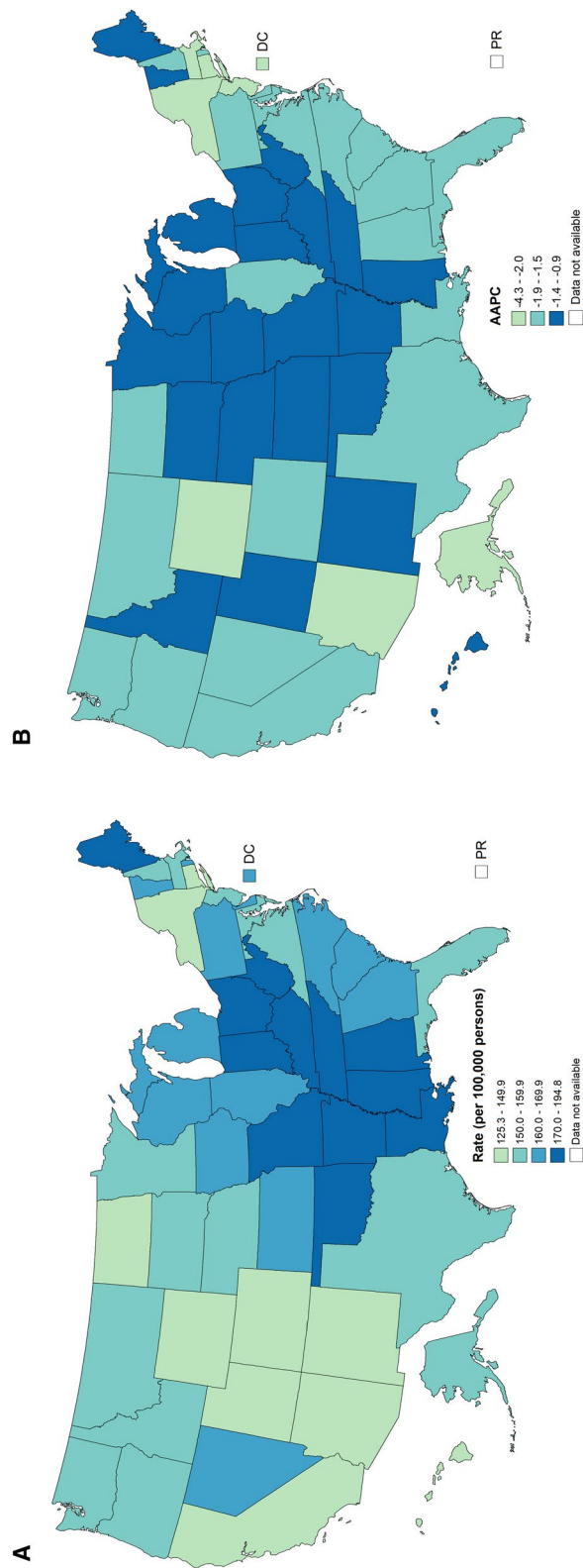


Figure 7. (A) Age-standardized overall cancer death rates for 2013 through 2017 and (B) the average annual percent change (AAPC) in rates for 2013 through 2017 are illustrated by state. The AAPC was a weighted average of the annual percent changes (APCs) over the fixed 5-year interval from 2013 to 2017 using the underlying joinpoint regression model, which allowed up to 4 different APCs, for the 17-year period from 2001 to 2017. Stable AAPCs were not statistically significantly different from zero ($P < .05$). DC indicates District of Columbia; PR, Puerto Rico.

group of cancers may be because of increases in sites associated with human papillomavirus, such as tonsil, base of tongue, and oropharynx.³⁸

Changing trends in prostate cancer incidence and mortality largely reflect changes in the US Preventive Services Task Force (USPSTF) recommendations for prostate-specific antigen (PSA) screening.^{39,40} In 2008, the USPSTF considered evidence to be insufficient for recommending routine PSA screening in men aged 50 to 74 years and recommended against PSA screening in men aged ≥ 75 years,⁴¹ and, in 2012, subsequently recommended against routine PSA screening in men of any age.⁴² Subsequent declines in PSA test use have been associated with declines in prostate cancer incidence during 2009 through 2014.³⁹ Recent stabilization in prostate cancer mortality trends may be related to an increasing incidence of distant-stage disease since 2008.^{39,43} The USPSTF issued new prostate cancer screening recommendations in 2018, which recommend that men aged 55 to 69 years talk with their clinician about the benefits and harms of prostate cancer screening with the PSA test.⁴⁴

There are some trends in the positive direction. For the first time in decades, incidence rates for thyroid cancer in the most recent 5-year period stabilized among both males and females overall. Increasing trends in thyroid cancer incidence over the past several decades have been attributed in part to increasing use of ultrasound-guided fine-needle aspiration biopsy,⁴⁵ and a slowing of these trends, beginning in 2009, has been attributed to revisions in American Thyroid Association management guidelines for thyroid nodules, which recommended against biopsy of nodules < 1 cm in the absence of certain high-risk characteristics.⁴⁶ Recent changes in thyroid cancer guidelines are intended to reduce overdiagnosis and overtreatment of relatively low-risk tumors, which can result in long-term decrement in health-related quality of life and financial burden.^{47,48} Although thyroid cancer incidence rates stabilized among males and females of all ages, rates continued to increase among AYAs. An analysis of differentiated thyroid cancer incidence among children and AYAs found increasing trends for large as well as small tumors.⁴⁹ Changes in prevalence of several risk factors for thyroid cancer, including low-dose radiation exposure⁵⁰ and excess body weight,⁵¹ may contribute to increasing incidence rates in more recent birth cohorts.

Recent rapid declines in death rates from melanoma of the skin, first noted in last year's report,¹ likely have resulted from the introduction of new therapies that have improved survival rates for advanced melanoma.⁵²

New therapies, including targeted and immune checkpoint inhibitors, have increased the median survival for advanced-stage melanoma from approximately 9 months to at least 2 years, based on clinical trials. Early indications of the new therapies' effect on population-level survival were observed in a study using the SEER-18 database. The study found that 2-year survival among patients diagnosed with metastatic melanoma in the post-targeted era (2011-2014) was 28.3%, which is a statistically significant improvement over the 23.5% observed in the pre-targeted era (2004-2010).⁵³

Accelerating rates of declines in death rates from lung cancer, the most common cause of cancer death in both men and women, may reflect increases in survival related to improved treatments, improved access to diagnostic and treatment services, and possibly earlier diagnosis because of implementation of low-dose computed tomography screening in high-risk populations.⁵⁴ The selection of individualized lung cancer treatment can be based on characteristics such as disease stage, histologic features, and the presence of treatable oncogenic alterations.⁵⁵⁻⁵⁸

Because of the growth and aging of the US population, the number of cases and deaths for many cancers continue to increase even while the age-standardized incidence and death rates decrease.^{59,60} For example, during 2001 through 2017, the number of cancer deaths increased from 553,760 to 599,108, despite death rates decreasing from 196 to 153 deaths per 100,000 persons.⁷ Similarly, the number of malignant cancers reported increased from 1.4 million in 2001 to 1.7 million in 2016, despite the incidence rate declining 0.6% per year.⁶¹ Future trends in cancer incidence and mortality will be affected by changes in the demographics of the US population, which is projected to be older and more racially and ethnically diverse.⁶² Tailored approaches for proven, evidence-based interventions may help maintain progress in decreasing the cancer burden in the United States.²

Strengths and Limitations

The major strength of this study is the high population coverage of cancer incidence data for the United States and Puerto Rico (99% for incidence rates during 2012-2016 and 92% for incidence trends during 2001-2016). Data from some cancer registries did not meet criteria for data quality or completeness for some years; however, because of the high population coverage, these exclusions likely do not materially impact national trends. Another strength is that this is the first Annual Report that has examined race and ethnicity in mutually exclusive groups. Previous reports categorized

race as 4 nonoverlapping groups, regardless of ethnicity, and categorized ethnicity as 2 nonoverlapping groups, regardless of race. Mutually exclusive groups allow more direct comparisons between categories. In addition, this report restricts the analysis of incidence data to cases categorized by the third edition of the ICD-O, so rates and trends may not be comparable to previous reports. Many of this study's limitations, including potential lack of representativeness of cancer incidence data of the US population, have been described in previous reports.^{37,63-65} In some instances, this report groups cancers that are anatomically close to each other but may have different etiologies (such as cancers of the oral cavity and pharynx) or cancers at 1 site with different histologic types (such as small-cell and non-small cell lung cancer) or molecular types (such as breast cancer); more detailed examinations of these cancers are beyond the scope of this report.

Conclusions

Although overall cancer death rates continue to decline, incidence rates are leveling off among males and increasing slightly among females. These trends reflect population changes in cancer risk factors, screening test use, diagnostic practices, and treatment advances. Many cancers can be prevented or treated effectively if found early. Population-based cancer incidence and mortality data can be used to inform efforts to decrease the cancer burden in the United States and regularly monitor progress toward goals.

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